



SENSORS AND SWITCHES

for pressure, temperature, level and flow.

All the instruments meet the guidelines of the European Community (EU).
It is confirmed that these products are approved acc. to following standards.



DIN/EN 61000-6-2
DIN/EN 61000-6-3



Note!

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Technical subject to change. February 2026.

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Product overview

Measurement

Pressure and temperature sensors

SCP03	SCP04	SCP08
		
Pressure sensor for mobile and industrial applications	Pressure transmitter for hydrogen applications	Pressure sensor for press construction and die-casting
Page 10-13	Page 15-19	Page 20-21

SCP09	SCP10	SCP11
		
Pressure sensor for industrial applications in the mid and high pressure ranges	For explosive environment with ATEX	Pressure Transmitter for hydrogen applications with ATEX approval
Page 22-26	Page 27-31	Page 32-35

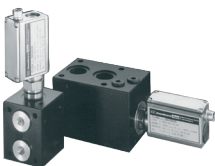
SCPSi



Pressure switch with IO-Link

Page 36-38

Volumetric flow rate sensors

SCQ	SCFT	SCVF
		
For quick flow changes Measures in both directions	Low loss measuring of volume flow	Measures different substances Measures lower volume flows (leakage measurements)
Page 46-49	Page 50-53	Page 54-59

Product overview

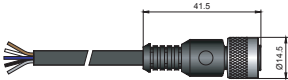
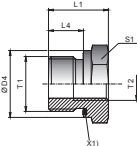
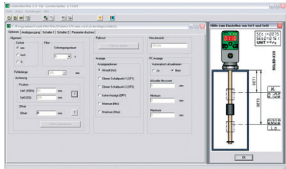
Measurement, display and switching

I/O-Link Switch
Family

SCPSDI	SCTSDI	SCLSDI
		
Pressure display and monitoring	Temperature display and level monitoring	Level display and monitoring
Page 62-65	Page 66-69	Page 74-78

SCLTSDI	SCTSDI Modular
	
Level/temperature display and monitoring	
Page 79-82	Page 70-73

Accessories

SCK cable	SCA adapter	Software ControllerWIN
		
Page 83-84	Page 85-86	Page 87-88

Selection guide pressure sensors

		SCP03	SCP04	SCP08	SCP09
Pressure-range	0...bar / (psi) relative	04...1000 (58...14,504)	04...1000 (58...14,504)	250/400/600/1000 (3626...14,504)	10...600 (145...8702)
	-1...bar / -14.5 (psi) relative	3...24 (43,5...348)			
	0...bar / (psi) absolut				
Order qty.		50 pcs	50 pcs	1 / 5 / 50 pcs	
Accuracy		0,5 %	0,5 %	0,5 %	0,5 %
Display					
Output	Switching Output				
	IO-Link				
	0,5...4,5 V (ratiometric 5V)	•	•		•
	0,5...4,5 V (nominal 24V)	•			•
	0...5 V	•			•
	1...6 V	•			
	0...10 V	•	•	•	•
	0...20 mA	•			
	4...20 mA (3-wire)	•			
	4...20 mA (2-wire)	•	•	•	•
	CAN				
Electrical Plug	M12	•	•	•	•
	DIN EN 175301-803 Form A	•	•	•	•
	DIN Micro 9.4				
	AMP Superseal	•			
	Deutsch DT04 4-pin	•			
	Deutsch DT04 3-pin	•	•		
	Junior Timer	•			
	Cable 2m	•			
Thread	G1/4 BSPP ED	•	•	•	•
	G 1/4 O-ring	•			
	1/4 NPT	•	•		•
	7/16-20 UNF	•	•		•
	9/16-20 UNF	•			
Wetted parts	Stainless steel/ Soft sealing	FKM		FKM	FKM
	Stainless steel/ Metall sealing		•		
Approvals	CE	•		•	•
	Marine				
	Safety SIL / PL				

Pressure sensors

Device features

- Long-term stability
- Immune to interference
- Rugged design
- Reliable



SensoControl® sensors feature long-term stability, interference immunity, a sturdy high-quality construction and a wide range of variants.

The sensors are designed and manufactured in our own production facilities under established standards for the industrial instrumentation and control systems. This allows us to easily adapt them to customer requirements or to critical applications.

We carefully consider the special requirements for automation and mobile hydraulics during the design phase. So our **SensoControl®** sensors are ideally suitable for the permanent series use in industrial and mobile applications.

Pressure sensors

The housing and all parts of the pressure sensors that touch the substances are manufactured from stainless steel. This provides a large range of media tolerability. A wide range of applications is possible due to the combination of high interference immunity and high resistance to external influences (shock, vibration and temperature).

The application areas are varied: from process engineering test rigs, conveying and lifting equipment, mobile hydraulics, general machine construction, pneumatic construction and hydraulic plant construction.

The SCP should be used when the pressure needs to be monitored reliably for long periods.

In this case the optimal sensor type can be selected from different product series according to the needs of the application. Different connecting plugs, output signals and connection threads are also available.

Pressure sensors - Overview

SCP03



SCP04



SCP08



Range of use	Pressure sensor for mobile and industrial applications - Up to 1000 bar - G1/4 DIN 3852-11 (E) - Compact design - Long term stability - Wide temperature range -40...125°C	Pressure sensor for hydrogen applications - Stainless steel measuring cell - Small design - Stainless steel housing - Up to 1000 bar - EC79/2009 (up to 600 bar) - High protection degree - Resistant to shock and vibration	Pressure sensor for press construction and die-casting 250 / 400 / 600 / 1000 bar G1/4" 0-10 V / 4...20 mA 2-wire - M12x1 / DIN - Reinforced internal design - Persistence against shock & vibration - Made for high pressure acceleration - High dynamic signal
Application	- Mobile hydraulic - Transport vehicles - Conveyor vehicles - Commercial vehicles - Automotive technology - Brake systems - Oil pressure - Test equipment and technology - Gearbox control	Hydrogen applications	- Press construction - Die-casting
Order code	SCP03-xxx-xx-xx	SCP04-xxx-xx-0xQ8	SCP08-xxxx-x4-0x
Refer to page	10-13	15-19	20-21

SCPSi



Range of use	IO-Link Pressure sensor or switch - Pressure sensor / -switch - Temperature measurement - Industry 4.0-ready - IO-Link 1.1 - Smart Sensor Profile 2nd edition - Plug & Play - Compact - Optimized design - Adjustable via IO-link - Readable via IO-Link - Useable as IO-Link sensor or switch - Monolithic pressure cell
Application	- Injection-mould machines - Tool-making machines - Power packs - Special machine construction - Replacement for mechanical pressure switches
Order code	SCPSi-xxx-04-07
Refer to page	29-31

Pressure sensors - Overview

	SCP09	SCP10	SCP11
			
Range of use	Pressure sensor for industrial applications, inc. injection molding - Up to 600 bar - G1/4 DIN 3852-11 (E)" 0-10 V / 4...20 mA 2-wire - M12x1 / DIN - Reinforced internal design - Extreme persistence against shock & vibration - Made for mid and high pressure acceleration	Pressure transmitter for use with liquid and gaseous media in hazardous areas - Up to 1000 bar - Monolithic design - High media compatibility - Negative pressure resistant - ATEX approved	Pressure transmitter for chemical and physical requirements of hydrogen - For Hydrogen - Optimized media connections - High media compatibility - EC79/2009 (up to 600 bar) - Negative pressure resistant - Monolithic design
Application	- Hydraulics and pneumatics - Mobile hydraulics and Off-High-way Vehicles - Pumps and Compressors - Air Conditioning and Refrigeration Systems - Plant Engineering and Automation	- For explosive environment - ATEX approved - Universal applicable - Mobile hydraulic - Industrial hydraulic - Off and onshore	Hydrogen application with ATEX approval
Order code	SCP09-xxx-xx-xx	SCP10-xxx-xx-xx	SCP11-xxx-xx-xx
Refer to page	22-26	27-31	32-35

SCP03 pressure sensor

Device features

- Monolithic design
 - No internal seal
 - No material mix
 - No weld seam
- High media compatibility
- Measuring range from -1 to 1000 bar / -14.5 to 14,504 psi
- Negative pressure resistant
- Many connections



The SCP03 is a pressure sensor for liquid and gaseous media.

The digitally calibrated piezoresistive measuring cell detects negative pressures from -1 bar up to high pressures of 1000 bar.

The pressure connection in contact with the medium has a monolithic design. This eliminates the need for internal seals and weld seams. A mix of materials is avoided.

The resulting low permeability in combination with the stainless steel results in broad media resistance.

The compact stainless-steel housing allows space-saving use, even in harsh environmental conditions. With its wide range of pressure ranges, output signals and connectors, the SCP03 can be used in industrial and mobile applications.

The packaging variant optimized for OEM's is environmentally friendly, cost-optimized and facilitates handling.

Typical application range

- Mobile hydraulics
- Transport vehicles
- Conveyor vehicles
- Commercial vehicles
- Automotive technology
- Brake systems
- Oil pressure
- Test equipment and technology
- Gearbox control

SCP03 pressure sensor

Technical data

SCP03-	004R	010R	010R	025R
Pressure range -1 ... bar P _n relative (-14.5 ... psi)	3 (43,5)	9 (130)	15 (218)	24 (348)

SCP03-	004	010	016	025	035	040	060	100	250	400	500	600	1000
Pressure range P _n relative 0 ... bar / (psi)	4 (58)	10 (145)	16 (232)	25 (363)	35 (500)	40 (580)	60 (870)	100 (1450)	250 (3626)	400 (5800)	500 (7300)	600 (8702)	1000 (14,504)
Overload pressure P _{max} DIN EN 60770-1 (bar) relative	2 x P _n												1,4 x P _n
Burst pressure P _{burst} DIN EN 60770-1 (bar) relative	3 x P _n												2 x P _n

SCP03-	0150P	0250P	1000P	3000P	5000P	9000P
Pressure range P _n relative 0... (psi)	150	250	1000	3000	5000	9000
Overload pressure* P _{max}	2 x P _n					
Burst pressure** P _{burst}	3 x P _n					

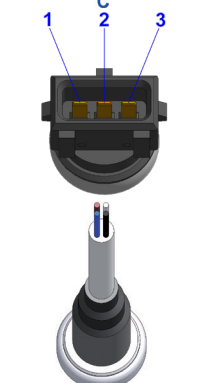
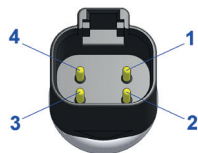
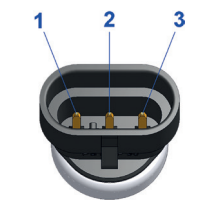
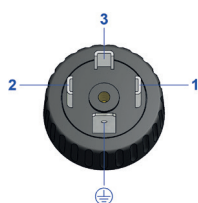
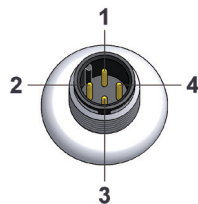
General		
Response time	≤1 ms	
Load change	> 100 million	
Material Housing	EN/DIN 1.4301	
Material Electr. Connector	PBT-GF30 black	
Weight	Approx. 80 g	
Accuracy parameter		
Non-linearity + Hysteresis + Repeatability	≤0.35 %FS	
Long-term stability	≤ ± 0.25 %FS / year	
Overall Accuracy		
	< 10 bar (145 psi)	≥ 10 bar (145 psi)
@ 25°C	≤ 0.5 %FS	≤ 0.5 %FS
-40°C bis 105°C	+ -2,5%	
@ 0°C...+85°C	≤ 2 %FS	≤ 1 %FS

Ambient conditions	
Media temperature	-40...+125 °C / (-40...257°F)
Operation / Ambient temperature	-40...+105 °C / (-40...221°F)
Storage temperature	-40...+125 °C / (-40...257°F)
Vibration resistance	IEC 60068-2-6: 20 g
Shock resistance	IEC 60068-2-27: 1000 g
Conformity	
CE	EN 61326-1 EN61326-3-1
RoHS	Yes
MTTFd	> 100 years

Process connection	Torque	Seal	Wetted parts
G1/4A BSPP; DIN 3852 T11, Form E	25 Nm	DIN 3869-14-FKM	EN/DIN 1.4404 / FKM
SAE-4: 7/16-20 UNF O-ring	25 Nm	FKM	EN/DIN 1.4404 / FKM
SAE 6: 9/16-18 UNF O-ring	25 Nm	FKM	EN/DIN 1.4404 / FKM
G1/4 DIN ISO 228-1 O-ring	25 Nm	FKM	EN/DIN 1.4404 / FKM
1/4 NPT	25 Nm		EN/DIN 1.4404

SCP03 pressure sensor

Pin assignment



Output signal	(2 wire) 4...20 mA	0...20 mA 4...20 mA	0.5...4.5 V 0...5 V	1...6 V 0...10 V	0.5...4.5 V ratio.
Supply Voltage V ₊	10...32 VDC	12...32 VDC	8...32 VDC	12...32 VDC	5 V ±10%
Load _{max}	≤ (V ₊ - 10V) / 20 mA [kΩ]		4.7 [kΩ]		
Overvoltage	50 VDC				
Short circuit	Yes				
Reverse polarity	Yes				
Signal on GND / V ₊	Yes				
M12x1 4-pole					
Pin 1	V ₊				
Pin 2	P-Signal				
Pin 3	n.c.	0 V / GND			
Pin 4	n.c.	n.c.			
IP65 (for p ≥ 60 bar IP67)*					
DIN EN 175301-803 Form A 4-pole (old 43650)					
Pin 1	P-Signal				
Pin 2	n.c.	0 V / GND			
Pin 3	V ₊				
Pin 4 / GND	n.c.				
IP54					
AMP Superseal 1.5					
Pin 1	P-Signal	0 V / GND			
Pin 2	n.c.	P-Signal			
Pin 3	V ₊				
IP65 (for p ≥ 60 bar IP67)*					
DT04-4P					
Pin 1	V ₊				
Pin 2	P-Signal	0 V / GND			
Pin 3	n.c.	P-Signal			
Pin 4 / GND	n.c.				
IP65 (for p ≥ 60 bar IP67)*					
DT04-3P					
A	V ₊				
B	n.c.	P-Signal			
C	P-Signal	0 V / GND			
IP 67					
Junior Timer					
Pin 1	P-Signal	0 V / GND			
Pin 2	n.c.	P-Signal			
Pin 3	V ₊				
IP65 (for p ≥ 60 bar IP67)*					
Cable					
Bn	V ₊				
Black	P-Signal				
Blue	n.c.	0 V / GND			
IP 67					

*IP67 when plugged

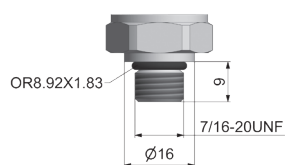
SCP03 pressure sensor

Pin assignment

Connectors & Ports

SCP03-...-x7

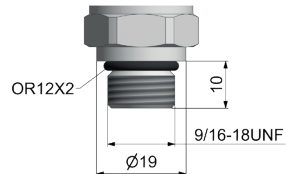
M12 4P



SCP03-...-x6

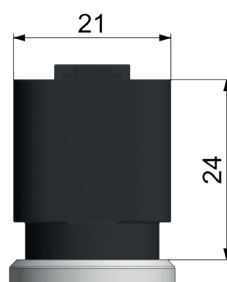
DIN EN 175301-803

Form A



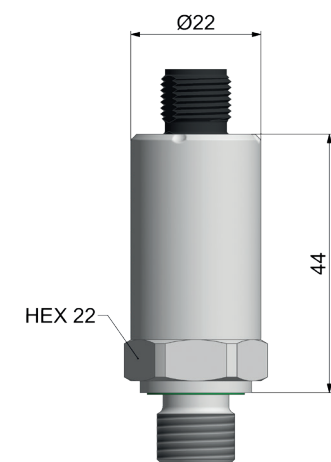
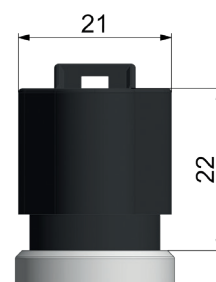
SCP03-...-xD

DT04 4P



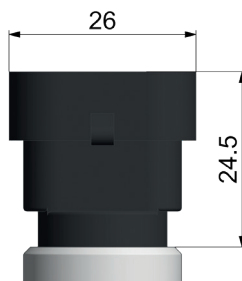
SCP03-...-xE

DT04 3P



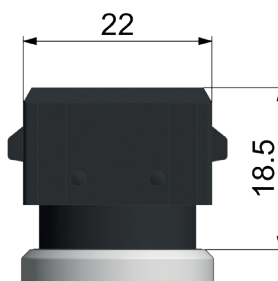
SCP03-...-xA

Superseal



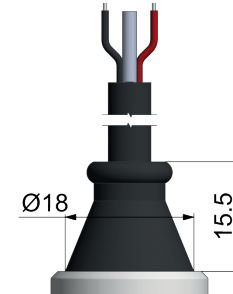
SCP03-...-xJ

Junior Timer 3P



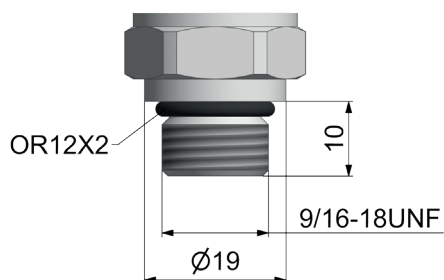
SCP03-...-x0

Kabel



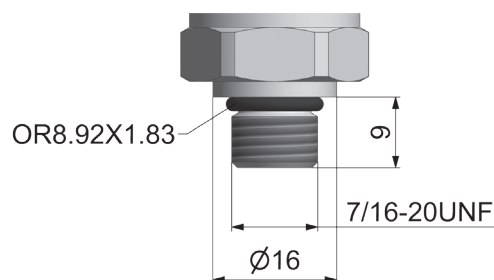
SCP03-xxx-x6-xx

SAE 06 - O-ring



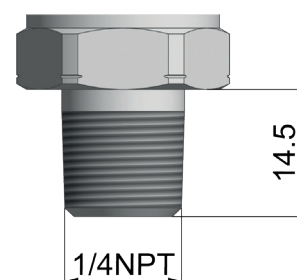
SCP03-xxx-x7-xx

SAE 04 - O-ring



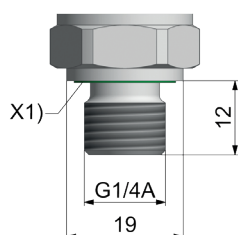
SCP03-xxx-x5-xx

1/4 NPT



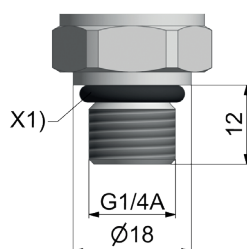
SCP03-xxx-x4-xx

G 1/4, DIN 3852 T 11 (Form E)



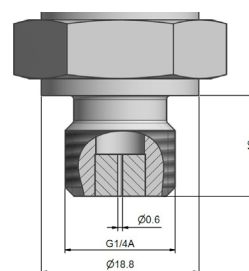
SCP03-xxx-x8-xx

G 1/4, O-ring



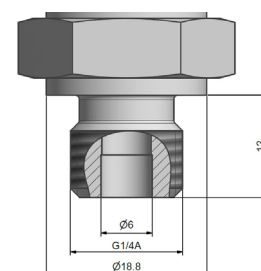
SCP03-xxx-xx-Dx

G 1/4, with damping



SCP03-xxx-xx-xx

G 1/4, without damping



X1) = ED-Seal

X1) = O-ring



SCP03 pressure sensor

Order code

Order quantity

Available single versions

Pressure sensor SCP03 Industrial SCP03-xxx-xx-Dx

Pressure range

0...10 bar	010
0...25 bar	025
0...60 bar	060
0...250 bar	250
0...400 bar	400
0...600 bar	600

Output signal

4...20 mA (3-wire)	2
4...20 mA (2-wire)	3
0...10 V	4

Process connection

G1/4 BSPP	4
-----------	---

With damping D

Connecting plug

Device connector DIN EN 175301-803 Form A 4-pole	6
Circular connector M12x1 4-pole	7

Pressure sensor SCP03 Mobile SCP03-xxx-xx-Dx

Pressure range

0...10 bar	010
0...25 bar	025
0...60 bar	060
0...250 bar	250
0...400 bar	400
0...600 bar	600

Output signal

4...20 mA (2-wire)	3
0.5...4.5 V (ratiometric)	R

Process connection

G1/4 BSPP	4
-----------	---

With damping D

Connecting plug

Device plug DT04 4 pole	D
-------------------------	---

Order example

150x SCP03-400-34-07Q8

150 Single sensors (multiple of 50's)

Pressure range 0...400 bar

Output signal 4 to 20 mA (2-wire)

G1/4 BSPP

Without damping

M12 connecting plug 4-pole



Pressure sensor SCP03

Pressure range

-1...3 bar	004R
-1...9 bar	010R
-1...15 bar	016R
-1...24 bar	025R
0...4 bar	004
0...10 bar	010
0...16 bar	016
0...25 bar	025
0...35 bar	035
0...60 bar	060
0...100 bar	100
0...160 bar	160
0...250 bar	250
0...400 bar	400
0...500 bar	500
0...600 bar	600
0...1000 bar	1000
0...150 psi	0150P
0...250 psi	0250P
0...1000 psi	1000P
0...3000 psi	3000P
0...5000 psi	5000P
0...9000 psi	9000P

Output signal

0...20 mA	1
4...20 mA (3-wire)	2
4...20 mA (2-wire)	3
0...10 V	4
0...5 V	A
1...6 V	B
0.5...4.5 V (ratiometric)	R
0.5...4.5 V (nom.)	S

Process connection

G1/4 BSPP	4
1/4 NPT (P _n max. = 600 bar)	5
9/16-18 UNF, SAE 6 O-ring (P _n max. = 400 bar)	6
7/16-20 UNF SAE-4 O-ring (P _n max. = 400 bar)	7
G1/4 O-ring (P _n max. = 600 bar)	8

Damping

Without damping	0
With damping	D

Connecting plug

Device connector DIN EN 175301-803 Form A 4-pole	6
Circular connector M12x1 4-pole	7
Stationary cable 2 m	0
Device plug AMP Superseal	A
Device plug DT04 4 pole	D
Device plug DT04 3 pole	E
Junior Timer 3-pole	J

Minimum order qty:

Q8: Multiple of 50 pcs.

SCP03-xxx-xx-xxQ8

SCP04 pressure sensor

Device features

- Monolithic design
 - No internal seal
 - No material mix
 - No weld seam
- EC79/2009 approved (up to 600 bar)
- High media compatibility (hydrogen)
- Up to 1000 bar (up to 14,504 psi)
- Negative pressure resistant
- Special connections



The SCP04 pressure sensor is designed to meet the chemical and physical requirements of hydrogen applications.

The digitally calibrated piezoresistive stainless steel measuring cell detects pressures up to 1000 bar. The connection to the sensing elements is made via a special bonding and thus remains stable even at low temperatures, shocks or vibrations.

The measuring cell and the pressure connection in contact with the medium are made in one piece. This eliminates the need for internal seals and weld seams. A mix of materials is avoided. The construction was designed to prevent embrittlement of the metal surface by ionized hydrogen.

The monolithic design eliminates leakage due to material fatigue at internal seals. The SCP04 has no pressure transfer fluid, no large pressurized areas, and is vacuum-tight and elastomer-free.

The resulting low permeability in combination with the stainless steel results in a wide media resistance. The process connections have been designed to be gasket-free for hydrogen applications.

The compact stainless steel housing allows space-saving use, even under harsh environmental conditions.

Typical application range

- Hydrogen applications

SCP04 pressure sensor

Technical data

SCP04-	004	025	400	500	600	1000
Pressure range P_n relative 0 ... bar / (psi)	4 (58)	25 (363)	400 (5800)	500 (7300)	600 (8702)	1000 (14,504)
Overload pressure P_{max} DIN EN 60770-1 (bar) relative	$2 \times P_n$					$1,4 \times P_n$
Burst pressure P_{burst} DIN EN 60770-1 (bar) relative	$3 \times P_n$					$2 \times P_n$

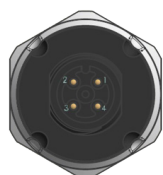
General		
Response time	≤1 ms	
Load change	> 10 million	
Material Housing	EN/DIN 1.4301	
Weight	Approx. 120 g	
MTTFd	> 100 years	
Accuracy parameter		
BFSL: Non-linearity, Zero-point error, Hysteresis, Repeatability	≤0.35 %FS	
Limit-point setting	≤0.5 %FS	
Resolution	FS/1500 digit	
Long-term stability	≤0.3 %FS / year	
Overall Accuracy		
	< 10 bar (145 psi)	≥ 10 bar (145 psi)
@ 25°C	≤0.5 %FS	≤0.5 %FS
@ -5°C...+85°C	≤2 %FS	≤1.5 %FS
@ -40°C...+105°C	≤3 %FS	≤2 %FS

Ambient conditions	
Media temperature	-40...+125 °C / (-40...257°F)
Operation / Ambient temperature	-40...+105 °C / (-40...221°F)
Storage temperature	-40...+125 °C / (-40...257°F)
Vibration resistance	IEC 60068-2-6: 20 g
Shock resistance	IEC 60068-2-27: 100 g
Conformity	
CE	EN 61326-1 EN 61326-2-3
RoHs	2011/65/EU

Process connection	Torque	Wetted parts
7/16"-20UNF-2A	25 Nm	316L; EN/DIN 1.4404 / FKM
9/16"-18UNF-2A	35 Nm	316L; EN/DIN 1.4404 / FKM
9/16"-18UNF-2A / 59°	25 Nm	316L; EN/DIN 1.4404
G1/4 B (EN 837)	25 Nm	316L; EN/DIN 1.4404
1/4 NPT	25 Nm	316L; EN/DIN 1.4404

SCP04 pressure sensor

Pin assignment



M12x1 (S763)



DIN EN 175301-803
Form A



DT04-3P

Output signal	(2 wire) 4....20 mA	0...10 V	0.5...4.5 V ratio.
Supply Voltage V ₊	10...32 VDC	12...32 VDC	5 V ±10%
Load _{max}	≤ (V ₊ - 10V) / 20 mA [kΩ]		4,7 [kΩ]
Overvoltage	50 VDC		
Short circuit	Yes		
Rever polarity	Yes		
Signal on GND / V ₊	Yes		
M12x1 4-pole			
Pin 1	V ₊		
Pin 2	P-Signal		
Pin 3	n.c.	0 V / GND	
Pin 4	n.c.	n.c.	
IP65 (for p ≥ 50 bar IP67)*			
DIN EN 175301-803 Form A 4-pole (old 43650)			
Pin 1	P-Signal		
Pin 2	n.c.	0 V / GND	
Pin 3	V ₊		
Pin 4 / GND	n.c.		
IP 54			
DT04-3P			
A	V ₊		
B	n.c.	P-Signal	
C	P-Signal	0 V / GND	
IP 67			

*IP67 when plugged

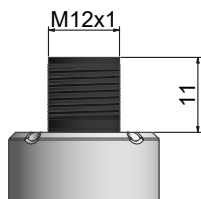
SCP04 pressure sensor

Pin assignment

Connectors & Ports

SCP04-...-07

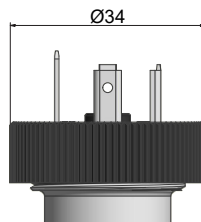
M12 4P



SCP04-...-06

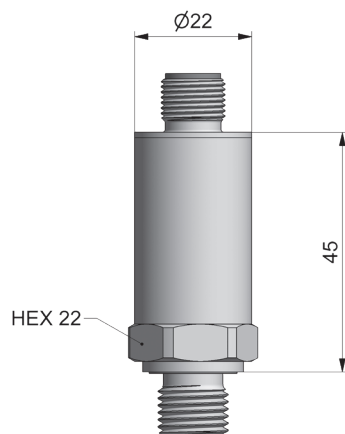
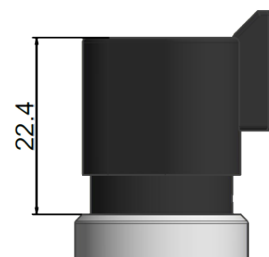
DIN EN 175301-803

Form A



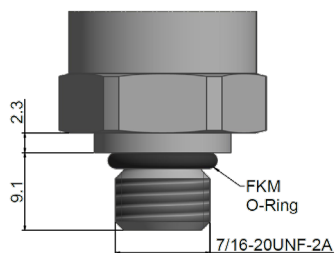
SCP04-...-0E

DT04 3P



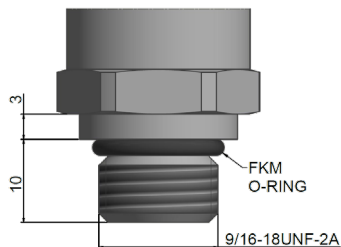
SCP04-xxx-x7-0x

7/16"-20UNF-2A



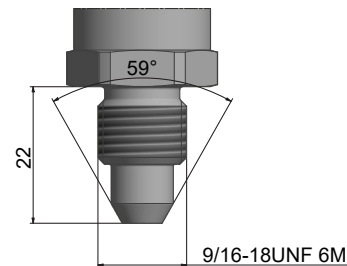
SCP04-xxx-x6-0x

9/16"-18UNF-2A



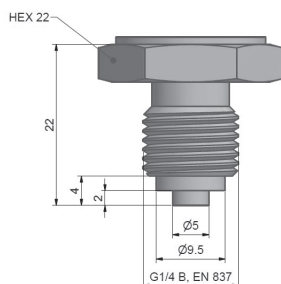
SCP04-xxx-x6M-0x

9/16"-18UNF-2M / 59°



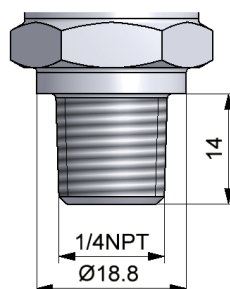
SCP04-xxx-xA-0x

G 1/4 B (EN 837)



SCP04-xxx-x5-0x

1/4 NPT



SCP04 pressure sensor

Order code

Pressure sensor SCP04	SCP04-xxx-xx-0xQ8
Pressure range (bar)	
0...4 bar	004
0...25 bar	025
0...400 bar	400
0...500 bar	500
0...600 bar	600
0...1000 bar*/**	1000
Output signal	
4...20 mA (2-wire)	3
0...10 V	4
0.5...4.5 V (ratiometric)	R
Process connection	
G1/4 B (EN 837)	A
1/4 NPT	5
7/16"-20UNF-2A	7
9/16"-18UNF-2A	6
9/16"-18UNF-2A / 59°	M
Connecting plug	
Device connector DIN EN 175301-803 Form A 4-pole	6
Circular connector M12x1 4-pole	7
Device plug DT04 3 pole	E
Minimum order qty:	
Q8: Multiple of 50 pcs.	

Additional Variances

Individual Pressure-ranges / calibration	available
Additional Ports	available
Individual Pin configuration	available
Brand label	available

*Up to EC79 600 bar approved

**Only thread 9/16"-18UNF-2M / 59°

Pressure sensor SCP08 for presses

Device features

- 250 / 400 / 600 / 1000 bar
(3,626 / 5,802 / 8,702 / 14,504 psi)
- G1/4"
- 0-10V / 4...20mA 2-wire
- M12x1 / DIN
- Reinforced internal design
- Persistence against shock & vibration
- Made for high pressure acceleration
- High dynamic signal



Particularly in die-casting applications the controlling for the piston requires a high dynamic pressure sensor. During this fast, high energetic process the components are stressed by shock, vibration and pressure acceleration.

The pressure sensor SCP08 measures the pressure via a special designed measurement cell and has a high adapted overload pressure to withstand the pressure peaks.

To avoid abrasion of the cell due to Diesel or similar effects, the process connection is protected by an adjusted drilling. The dimension of the drilling still guarantees an instantaneous pressure response.

To increase shock and vibration resistance, the relevant internal components are covered and reinforced. The speed of the sensor influences directly the quality of the production process.

The unique combination of accuracy, durability and high dynamic response makes the SCP08 ideal for the requirements of die-casting applications.

Typical applications

- Press construction
- Die-casting

Pressure and temperature sensors

Pressure sensor SCP09

Device features

- Measuring range
 - from 0-10 bar to 0-600 bar
(0-145 psi to 0-8700 psi)
- Wide range of
 - Ports
 - Connectors
 - Electrical outputs
- Stainless steel design
- Hermetic port
- Storage and operating media temperature
 - -40 to 125°C
- Operating ambient temp. range
 - -40 to 100°C
- Snubber option
- REACH/RoHS/CE/UKCA



The SCP09 pressure sensor is the ideal solution for customers with challenging measuring requirements for general industrial applications in the mid and high pressure ranges. The SCP09 features a wide range of ports, connectors, and analog electrical outputs for ease of integration in various industrial applications.

The SCP09's high quality stainless steel design features a hermetic port with no internal O-ring seals making it compatible with most media and suitable for harsh environments. With extreme shock and vibration capabilities, a wide operating temperature range, and high proof and burst pressures; the SCP09 is ideal for industrial applications including injection molding, CO₂, HVAC systems, and other hydraulic or pneumatic applications.

For dampening of pressure spikes due to hammer and cavitation a optional damping (Snubber) can be provided.

Typical applications

- Hydraulics and Pneumatics
- Mobile Hydraulics and Off-Highway Vehicles
- Pumps and Compressors
- Air Conditioning and Refrigeration Systems
- Plant Engineering and Automation

Pressure sensor SCP09

Technical data

SCP09-	010	016	025	040	050	060	100	160	200	250	350	400	500	600
Pressure range P_n relative 0 ... bar / (psi)	10 (145)	16 (232)	25 (363)	40 (580)	50 (725)	60 (870)	100 (1450)	160 (2321)	200 (2901)	250 (3626)	350 (5076)	400 (5802)	500 (7252)	600 (8702)
Overload pressure P_{max} DIN EN 60770-1 (bar) relative	60 (870)	60 (870)	60 (870)	200 (2901)	200 (2901)	200 (2901)	200 (2901)	500 (7252)	500 (7252)	500 (7252)	800 (11603)	800 (11603)	1200 (17405)	1200 (17405)
Burst pressure P_{burst} DIN EN 60770-1 (bar) relative	200 (2901)	200 (2901)	200 (2901)	2000 (29008)	2000 (29008)	2000 (29008)	2000 (29008)	2500 (36259)	2500 (36259)	2500 (36259)	4000 (58015)	4000 (58015)	4000 (58015)	4000 (58015)

General

Response time	≤2 ms
Durability	> 10 million

Accuracy parameter

Basic Accuracy IEC 61298-2 (Non-linearity + Hysteresis + zero & span)	± 0.5 %FS @ 25°C
BFSL Accuracy IEC 60770-1 (Non-linearity + Hysteresis + repeatability, excluding zero & span)	± 0.25 %FS @ 25°C
Non-linearity BFSL	± 0.2 % FS
Non-repeatability	± 0.1 % FS

Overall Accuracy

@ -20°C...+85°C	± 1.5 %FS
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Ambient conditions

Media temperature	-40...+125 °C / (-40...257°F)
Operation temperature	-40...+100 °C / (-40...212°F)
Storage temperature	-40...+125 °C / (-40...257°F)
Vibration resistance	IEC 60068-2-6: 30 g (10...2000Hz)
Shock resistance	IEC 60068-2-27: 500 g

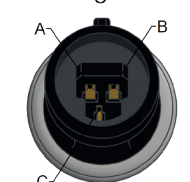
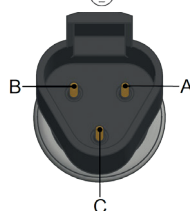
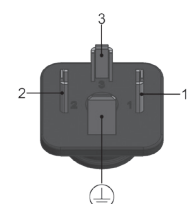
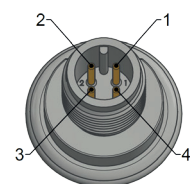
Conformity

CE Emission / Immunity	EN 61326
RoHs / REACH	Yes
UKCA	Yes
Radiated Immunity	100V/m @80...200MHz 200V/m @200...2700MHz

Process connection	Torque	Seal	Wetted parts
G1/4A DIN3852 T11, Form E	25 Nm	FKM	17-4PH EN/DIN 1.4542 ; FKM
7/16-20 UNF-2A J1926-3 O-ring	25 Nm	HNBR	17-4PH EN/DIN 1.4542 ; HNBR
1/4-19 PT (BSPT) (R1/4)	25 Nm	–	17-4PH EN/DIN 1.4542
1/4-18 NPTF	25 Nm	–	17-4PH EN/DIN 1.4542

Pressure sensor SCP09

Pin assignment

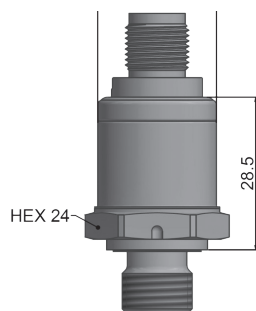


Output signal	(2 wire) 4...20 mA	0.5...4.5 V 0...5 V 1...5 V	0...10 V	0.5...4.5 V ratio.
Supply Voltage V_+	8...32 VDC	8...32 VDC	12...32 VDC	5 V $\pm$ 0.25 VDC
Overvoltage	36 VDC	36 VDC	36 VDC	–
Short circuit	–	Yes	Yes	Yes
Reverse polarity	Yes	Yes	Yes	Yes
Load _{max}	$\leq (V_+ - 8V) / 20mA$ [k Ω]	≥ 4.7 [k Ω]	≥ 4.7 [k Ω]	≥ 4.7 [k Ω]
M12x1 4-pole				
Pin 1	V_+	V_+	V_+	V_+
Pin 2	n.c.	n.c.	n.c.	n.c.
Pin 3	P-Signal	0 V / GND	0 V / GND	0 V / GND
Pin 4	n.c.	P-Signal	P-Signal	P-Signal
IP 67				
DIN EN 175301-803 Form A 4-pole (old 43650)				
Pin 1	V_+	V_+	V_+	V_+
Pin 2	P-Signal	0 V / GND	0 V / GND	0 V / GND
Pin 3	n.c.	P-Signal	P-Signal	P-Signal
Pin 4 / GND	n.c.	n.c.	n.c.	n.c.
IP 65				
DT04-3P				
A	V_+	V_+	V_+	V_+
B	P-Signal	0 V / GND	0 V / GND	0 V / GND
C	n.c.	P-Signal	P-Signal	P-Signal
IP 67				
Metri-Pack 150				
A	P-Signal	0 V / GND	0 V / GND	0 V / GND
B	V_+	V_+	V_+	V_+
C	n.c.	P-Signal	P-Signal	P-Signal
IP 67				

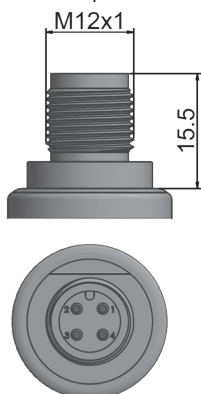
Pressure sensor SCP09

Pin assignment Connectors & Ports

SCP09-...-...-07
M12 4P



SCP09-...-...-07
M12x1 4-pole



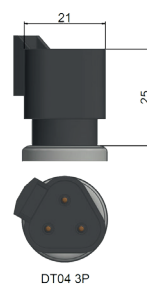
M12 4P

SCP09-...-...-06
DIN EN 175301-803 - Form A



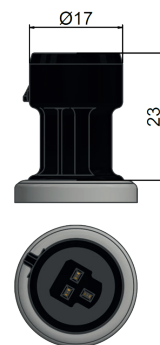
DIN EN 175301-803
FORM A

SCP09-...-...-0E
DT04 3P



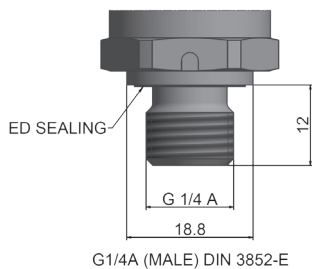
DT04 3P

SCP09-...-...-0P
Packard Metri Pack 150



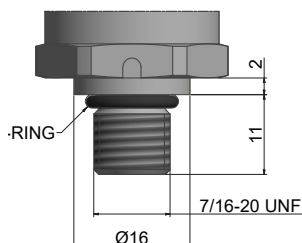
Packard Metri Pack 150

SCP09-xxx-x4-xx
G 1/4, DIN 3852 T 11 (Form E)



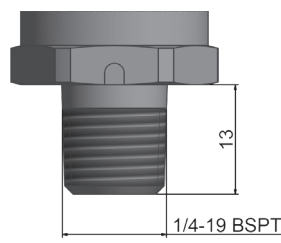
G1/4A (MALE) DIN 3852-E

SCP09-xxx-x7-xx
7/16-20 UNF-2A



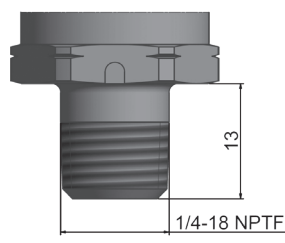
7/16-20 UNF J1926-3

SCP09-xxx-xR-xx
1/4-19PT (R1/4)



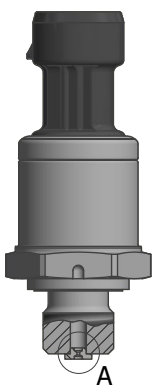
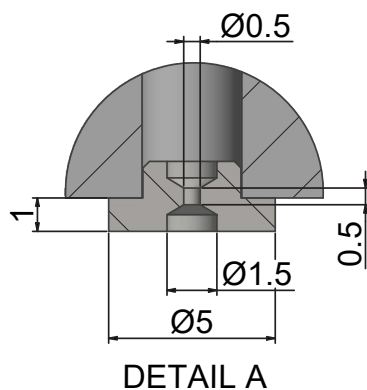
1/4-19 PT(BSPT) (R1/4)

SCP09-xxx-x5-xx
1/4-18NPT



1/4-18 NPTF

Optional with damping (Snubber)



Pressure sensor SCP09

Order code

Pressure sensor SCP09	SCP09-xxx-xx-xxQ
Pressure range	
0...10 bar	010
0...16 bar	016
0...25 bar	025
0...40 bar	040
0...50 bar	050
0...60 bar	060
0...100 bar	100
0...160 bar	160
0...250 bar	250
0...350 bar	350
0...400 bar	400
0...500 bar	500
0...600 bar	600
Output signal	
4...20 mA (2-wire)	3
0...10 V	4
0...5 V	A
1...5 V	D
0.5...4.5 V @ 5±0.25VDC (ratiometric)	R
0.5...4.5 V @ 8-32VDC (nom.)	S
Process connection	
G1/4A DIN 3852-E	4
1/4-18 NPT	5
7/16-20 UNF -2A	7
1/4 -19PT (R1/4)	R
Damping	
Without damping	0
With damping	D
Connecting plug	
Device connector DIN EN 175301-803 Form A 4-pole	6
Circular connector M12x1 4-pole	7
DT04 3 pole	E
Packard	P
Order quantity:	
Q8: Multiple of 50 pcs.	
Q10: Multiple of 500 pcs.	

Pressure transmitter SCP10

Device features

- ATEX
 - II 1G Ex ia IIC T4 Ga
 - II 2G Ex ia IIC T4 Gb
 - II 1G Ex ia IIB T4 Ga
- Monolithic design
 - No internal seal
 - No material mix
 - No weld seam
- High media compatibility
- Up to 1000 bar
- Negative pressure resistant



The intrinsically safe pressure sensor SCP10 has been specially developed for use with liquid and gaseous media in hazardous areas. It fulfills the requirements for conditions in:

Zone 0: II 1G Ex ia IIC T4 Ga

II 1G Ex ia IIB T4 Ga

Zone 1: II 2G Ex ia IIC T4 Gb

This sensor ensures maximum safety in critical environments.

The digitally calibrated piezoresistive stainless steel measuring cell detects pressures from 4 bar up to 1000 bar. The connection to the sensing elements is made via a special bonding and therefore remains stable even at low temperatures, shocks or vibrations.

The measuring cell and the pressure connection in contact with the medium are made from a single piece. This eliminates the need for internal seals and weld seams. A mix of materials is avoided.

Due to the monolithic design, leaks caused by material fatigue at joints are excluded by design.

The SCP10 has no pressure transmission fluid, no large pressurized surfaces and is absolutely vacuum-tight and elastomer-free.

The resulting low permeability in combination with the stainless steel results in broad media resistance.

The compact stainless steel housing enables space-saving use, even in harsh environmental conditions.

With the wide range of pressure ranges, output signals and connectors, the SCP10 can be used in industrial and mobile applications.

Pressure transmitter SCP10

Technical data

SCP10-	004	010	016	025	060	100	250	400	600	1000
Pressure range 0... bar P _n relative 0... (psi)	4 (580)	10 (145)	16 (232)	25 (362)	60 (870)	100 (1450)	250 (3625)	400 (5801)	600 (8702)	1000 (14503)
Overload pressure P _{max} (bar) relative	8	20	32	50	120	200	500	800	1200	1400
Burst pressure P _{burst} (bar) relative	12	30	48	75	180	500	1000	1400	1800	2000

Accuracy

Long-term stability	± 0.5 %FS @ 25°C	
Overall Accuracy	@ 25°C	≤ ± 0,5 %FS
	@ -5°C ...+85°C	≤ ± 1.5 %FS
	@ -40°C...+85 °C	≤ ± 2.5 %FS

General

Response time	≤2 ms
Pressure cycles	> 100 million
Material Housing	EN/DIN 1.4301
Weight	Approx. 100 g
RoHs	Yes
MTTFd	>100 years

ATEX

M12 metallic connector	II 1G Ex ia IIC T4 Ga	
Other connector	II 1G Ex ia IIB T4 Ga II 2G Ex ia IIC T4 Gb	
Certificate	IBExU24ATEX1042 EN IEC 60079-0 EN 60079-11	

Ambient conditions

Media temperature	-40 to 100 °C (-40 to 212 °F)
Operation / Ambient temperature	
Zone 0	-20 to 60 °C
Zone 1 / 2	-40 to 85 °C (-40 to 185 °F)
Storage temperature	-40 to 85 °C (-40 to 185 °F)
Vibration	IEC 60068-2-6: 20 g / 3 axes
Shock	IEC 60068-2-31 (1m free fall)
CE emission	EN 61326-1:2013 - section 7 EN 61326-2-3:2013
CE susceptibility	EN 61326-1:2013 - section 6 EN 61326-2-3:2013

Process connection	Torque	Seal	Wetted parts
7/16-20UNF-2A	25 Nm	FKM	316L (EN/DIN 1.4404) / FKM
9/16-18UNF-2A	35 Nm	FKM	316L (EN/DIN 1.4404) / FKM
G1/4 Form E	25 Nm	FKM	316L (EN/DIN 1.4404) / FKM
G1/4 O-ring	25 Nm	FKM	316L (EN/DIN 1.4404) / FKM
G1/4 B	25 Nm	none	316L (EN/DIN 1.4404)
¼ NPT	25 Nm	none	316L (EN/DIN 1.4404)

Pressure transmitter SCP10

Electrical Connection



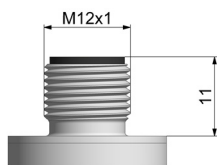
Output signal	4....20 mA (2wire)
Supply Voltage V ₊	20 ... 27 VDC
Load _{max}	< (V _{cc} -10 V) / 20 mA
Current consumption	3.6 ... 21.4 mA
Reverse polarity	Yes
Overvoltage protection	Yes
M12x1 4-pole	
Pin 1	V+
Pin 2	P-Signal
Pin 3	n.c.
Pin 4	n.c.
Protection Class (connected)	IP 67
ATEX	II 1G Ex ia IIC T4 Ga
DT04-3P	
A	V+
B	n.c.
C	P-Signal
Protection Class (connected)	IP 67
ATEX	II 2G Ex ia IIC T4 Gb
	II 1G Ex ia IIB T4 Ga
DIN EN 175301-803 Form A 4-pole	
1	P-Signal
2	n.c.
3	V+
GND	n.c.
Protection Class (connected)	IP 65
ATEX	II 2G Ex ia IIC T4 Gb
	II 1G Ex ia IIB T4 Ga
Conduit	
Red	V+
Black	P-Signal
Protection Class (connected)	IP68
ATEX	II 2G Ex ia IIC T4 Gb
	II 1G Ex ia IIB T4 Ga
Cable	
Red	V+
Black	P-Signal
Protection Class (connected)	IP 67
ATEX	II 2G Ex ia IIC T4 Gb
	II 1G Ex ia IIB T4 Ga

Pressure transmitter SCP10

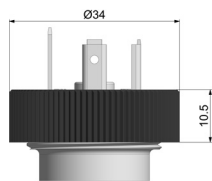
Drawings

M12x1

Metallic

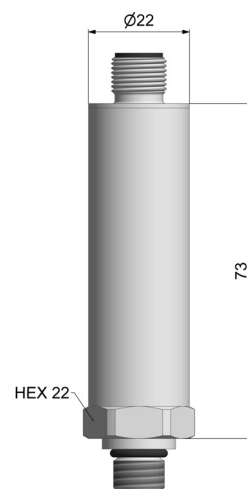
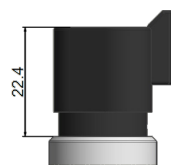


EN 175301

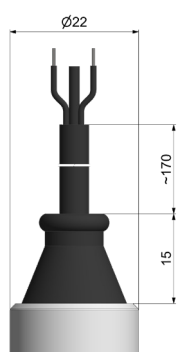


DT04

3pole

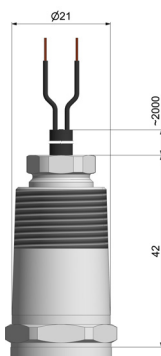


Cable (2m)



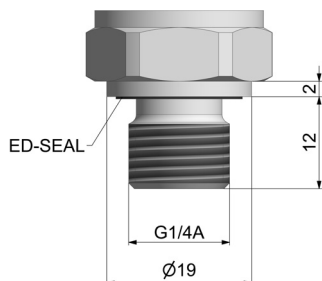
Conduit

1/2"-14 NPT (2m)

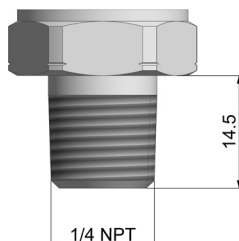


G1/4"A Form E

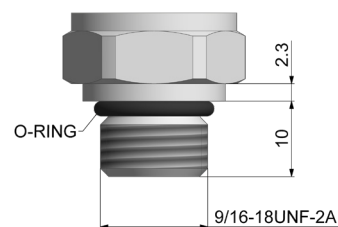
DIN EN ISO 1179-2



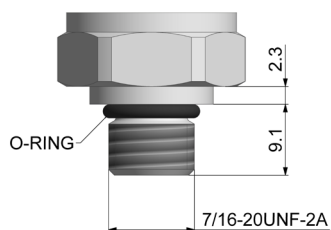
1/4"-18 NPT



9/16-18UNF-2A

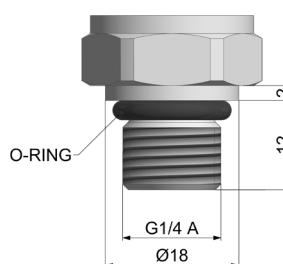


7/16-20UNF-2A



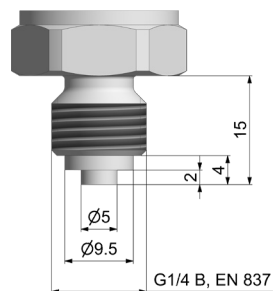
G1/4"A Form G

DIN EN ISO 1179-3

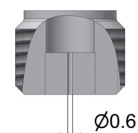


G1/4"B

EN 827



Snubber



Pressure transmitter SCP10

Order code

Pressure sensor SCP10	SCP10-xxx	-3x	-x x	-EXi0S
Pressure range				
0...4 bar	004			
0...10 bar	010			
0...16 bar	016			
0...25 bar	025			
0...60 bar	060			
0...100 bar	100			
0...250 bar	250			
0...400 bar	400			
0...600 bar	600			
0...1000 bar	1000			
Output signal				
4...20 mA (2-wire)		3		
Process connection				
G1/4"A DIN EN ISO 1179-2 Form E		4		
1/4-18 NPT		5		
9/16-18UNF-2A		6		
7/16-20UNF-2A		7		
G1/4"A DIN EN ISO 1179-3 Form G		8		
G1/4"B EN 827		A		
Damping				
Without damping			0	
With damping			D	
Connecting plug				
Cable 2m			0	
Cable 5m			005	
Cable 10m			010	
DIN EN 175301-803 Form A 4-pole			6	
M12x1 4-pole metallic			7	
DT04 3pole			E	
Conduit 1/2"-14 NPT (2m)			K	

Pressure transmitter SCP11

Device features

- ATEX
 - II 1G Ex ia IIC T4 Ga
 - II 2G Ex ia IIC T4 Gb
- Monolithic design
 - No internal seal
 - No material mix
 - No weld seam
- EC79/2009 (up to 600 bar)
- High media compatibility (hydrogen)
- Up to 1000 bar
- Negative pressure resistant
- Special connections



The intrinsically safe SCP11 pressure sensor is designed to meet the chemical and physical requirements of hydrogen applications. It fulfills the requirements for conditions in:

Zone 0: II 1G Ex ia IIC T4 Ga

Zone 1: II 2G Ex ia IIC T4 Gb

This sensor ensures maximum safety in critical environments.

The digitally calibrated piezoresistive stainless steel measuring cell detects pressures from 4 bar up to 1000 bar. The connection to the sensing elements is made via a special bonding and thus remains stable even at low temperatures, shocks or vibrations.

The measuring cell and the pressure connection in contact with the medium are made in one piece. This eliminates the need for internal seals and weld seams. A mix of materials is avoided. The construction was designed to prevent embrittlement of the metal surface by ionized hydrogen.

The monolithic design eliminates leakage due to material fatigue at internal seals.

The SCP11 has no pressure transfer fluid, no large pressurized areas, and is vacuum-tight and elastomer-free.

The resulting low permeability in combination with the stainless steel results in a wide media resistance. Specifically for hydrogen applications, a seal-free process connection has been implemented.

The compact stainless steel housing allows space-saving use, even under harsh environmental conditions.

Pressure transmitter SCP11

Technical data

SCP11-	004	025	400	500	600	1000
Pressure range 0... (bar) P _n relative 0 ... (psi)	4 (58)	25 (363)	400 (5800)	500 (7300)	600 (8702)	1000* ¹ (14503)
Overload pressure P _{max} (bar) relative	2 x P _n					1,4 x P _n
Burst pressure P _{burst} (bar) relative	3 x P _n					2 x P _n

*1 Only in combination with 9/16-18UNF-6M / 59°

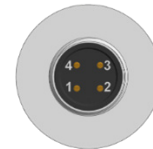
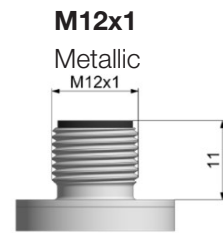
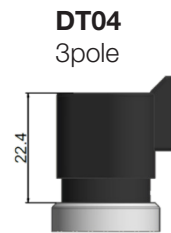
Accuracy	
Long-term stability	≤ ± 0.25 %FS / year
Overall Accuracy	@ 25°C ≤ ± 0,5 %FS
	@ -5°C ...+85°C ≤ ± 1.5 %FS
	@ -40°C...+85 °C ≤ ± 2,5 %FS
General	
Response time	≤2 ms
Pressure cycles	>100 million
Material Housing	EN/DIN 1.4301
Weight	Approx. 100 g
RoHs	Yes
MTTFd	>100 years
ATEX	
M12 metallic connector	II 1G Ex ia IIC T4 Ga
Other connector	II 2G Ex ia IIC T4 Gb
Certificate	IBExU24ATEX1042 EN IEC 60079-0 EN 60079-11
Ambient conditions	
Media temperature	-40 to 100 °C (-40 to 212 °F)
Operation / Ambient temperature	
Zone 0	-20 to 60 °C
Zone 1 / 2	-40 to 85 °C (-40 to 185 °F)
Storage temperature	-40 to 85 °C (-40 to 185 °F)
Vibration	IEC 60068-2-6 : 20 g / 3 axes
Shock	IEC 60068-2-31 (1m free fall)
CE emission	EN 61326-1:2013-section 7 EN 61326-2-3:2013
CE susceptibility	EN 61326-1:2013 - section 6 EN 61326-2-3:2013

Process connection	Torque	Seal	Wetted parts
7/16-20UNF-2A	25 Nm	HNBR	316L (EN/DIN 1.4404) / HNBR
9/16-18UNF-2A	35 Nm	HNBR	316L (EN/DIN 1.4404) / HNBR
9/16-18UNF 6M	25 Nm	none	316L (EN/DIN 1.4404)
1/4 NPT	25 NM	none	316L (EN/DIN 1.4404)

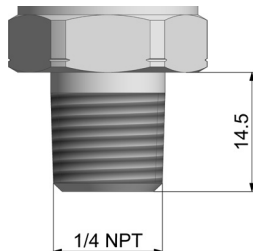
Pressure transmitter SCP11

Electrical Connection

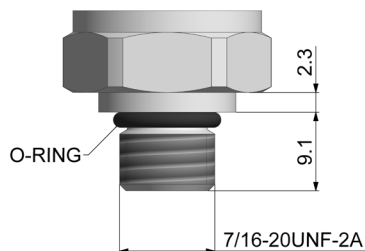
Output signal	4....20 mA (2wire)
Supply Voltage V_+	20 ... 27 VDC
Load _{max}	< ($V_{cc}-10\text{ V}$)/20 mA
Current consumption	3.6 ... 21.4 mA
Reverse polarity	Yes
Overvoltage protection	Yes
M12x1 4-pole	
Pin 1	V+
Pin 2	P-Signal
Pin 3	n.c.
Pin 4	n.c.
Protection Class (connected)	IP 67
ATEX	II 1G Ex ia IIC T4 Ga
DT04-3P	
A	V+
B	n.c.
C	P-Signal
Protection Class (connected)	IP 67
ATEX	II 2G Ex ia IIC T4 Gb



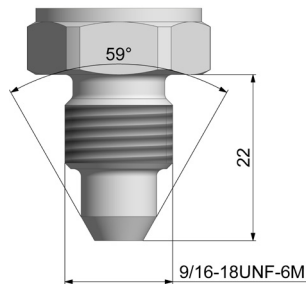
1/4"-18 NPT



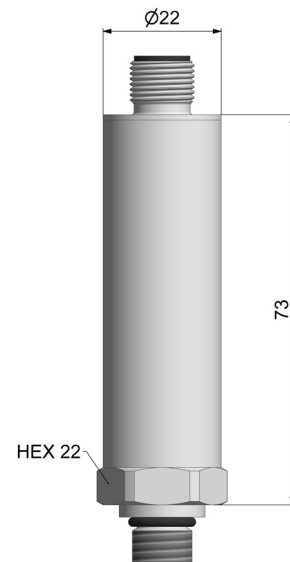
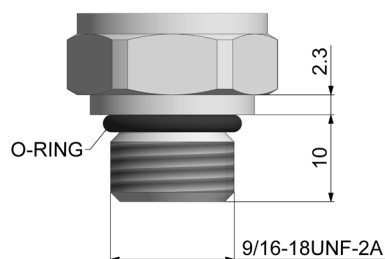
7/16-20UNF-2A



9/16-18UNF-6M



9/16-18UNF-2A



Pressure transmitter SCP11

Order code

Pressure sensor SCP11	SCP11-xxx	-3x	0 x	- EXi0S
Pressure range				
0...4 bar	004			
0...25 bar	025			
0...400 bar	400			
0...600 bar	600			
0...1000 bar*1	1000			
Output signal				
4...20 mA (2-wire)		3		
Process connection				
1/4"-18 NPT			5	
7/16-20UNF-2A			7	
9/16-18UNF-2A			6	
9/16-18UNF-6M / 59°			M	
Connecting plug				
M12x1 4-pole metallic				7
DT04 3pole				E

*1 Only in combination with 9/16-18UNF-6M / 59°

SCPSi pressure switch

Device features

- Pressure sensor / -switch
- Temperature measurement
- Industry 4.0-ready
- IO-Link 1.1
- Smart Sensor Profile 2nd edition
- Plug & Play
- Compact
- Optimized design
- Adjustable via IO-Link
- Readable via IO-Link
- Useable as IO-Link sensor or switch
- Monolithic pressure cell



The fully electronic pressure switch SCPSi is adjustable and free from susceptible mechanical and moving components.

With its digital interface and smart functions, the SCPSi is future-proof for the increasing demands of automation solutions.

The 2 switching outputs are individually and safely parameterized from the machine control system via the standardized digital IO-Link interface (IEC 61131-9). This replaces manual programming and the commissioning phase is considerably shortened. Devices can be replaced during operation without the need for reparameterization. In order to react promptly to machine status changes or process adjustments, the re-parameterization is carried out during operation.

As an alternative to the switching functions, diagnostic values, process data and status messages are recorded directly via IO-Link and enable subsequent more complex analyses. Via the integrated temperature measurement of the pressure measuring cell, the media or ambient temperature is recorded.

IO-Link replaces time-consuming manual programming and eliminates the need for a sensitive key display with the manufacturer-dependent setting menu. This more compact, more resistant design without key display, in combi-

nation with the smart functions & setting options, opens up new possibilities in machine design for the machine designer, with considerable savings potential.

The compact stainless steel housing allows space-saving use, even in harsh environments.

The proven stainless steel measuring cell with the wide pressure range (from -1 up to 600 bar) allows a wide range of applications for liquid and gaseous media. The media-contacting pressure connection with the pressure measuring cell is monolithically manufactured from a stainless steel without welds and sets new standards in media compatibility and pressure resistance.

The packaging variant optimized for OEM's is environmentally friendly, cost-optimized and facilitates handling.

Application examples

- **Injection-mould machines**
- **Tool-making machines**
- **Power packs**
- **Special machine construction**
- **Replacement for mechanical pressure switches**

SCPSi pressure switch

Technical data

SCPSi		001	004	010	025	060	100	250	400	600
Pressure range P _n	bar	-1...1	-1...4	-1...10	-1...25	0...60	0...100	0...250	0...400	0...600
vacuum tight / relative P _n	(psi)	(-14...14)	(-14...58)	(14...145)	(-14...362)	(0...870)	(0...1450)	(0...3625)	(0...5801)	(0...8702)
Overload pressure	bar	6	10	030	80	200	300	750	1200	1400
relative P _{max}	(psi)	(87)	(145)	(435)	(1160)	(2900)	(4351)	(10877)	(17404)	(20305)
Burst pressure	bar	9	15	100	150	500	800	1000	2000	2200
relative P _{burst}	(psi)	(130)	(217)	(1450)	(2175)	(7251)	(11603)	(14504)	(29007)	(31908)
Wetted parts		1.4542 (17-4PH); 1.4548; FKM		Monolitisch 316L; FKM						
Set point SP Range		1 - 100 %								
Reset point rP Range		0 - 99 %								
Steps / Incremental	mbar	0,1	1	1	1	10	10	10	100	100
Smallest hysteresis (SP-rP) & (FH-FL)	bar	0,001	0,01	0,01	0,01	0,1	0,1	0,1	1	1

General

Overall Accuracy @ RT [°1]	≤ 0,5 %FS
Min. pressure cycles	> 100 million
Material housing	Stainless steel 1.4404
Weight	approx. 80 g

Conformity

RoHS	2011/65/EU, 2015/863
CE	Yes
UKCA	Yes

Process connection

Thread	G1/4, DIN 3852 T11 (E)
Seal	ED type: FKM
Installation torque	Max. 35 Nm

Ambient conditions

Media temperature	-25 to 85 °C (-13 to 185°F)
Operation / Ambient temperature	-25 to 85 °C (-13 to 185°F)
Storage temperature	-40 to 85 °C (-40 to 185°F)
Vibration	DIN EN 60068-2-6, 20 g
Shock	DIN EN 60068-2-27, 500 g
MTTFd	>100 year

Accuracy

@ -40°C...-25°C	≤ 2,5 %FS
@ -25...0°C	≤ 1,5 %FS
@ 0...85°C	≤ 1 %FS

Temperature signal

Output	Via IO-Link
Short circuit	-40 to 125 °C
Resolution	1 K
Accuracy	± 10°K
t _{0,9}	80 sek.

Protection

Overvoltage	70 V
Short circuit	yes
Reverse polarity	yes
Signal on GND/V ₊	yes

Factory setting

SP1 / rP1	40 / 60% FS; Hno
SP2 / rP2	30 / 70% FS; Hno

Electronic Connectivity

Power supply voltage V ₍₊₎	18...30VDC
Connector	M12
Consumption	< 15 mA @ 24V
Output	2 switching outputs, NPN / PNP, 1 IO-Link output
Switch current	Max. 200mA
Max. switch frequency	200 Hz
Response time	≥ 3 ms

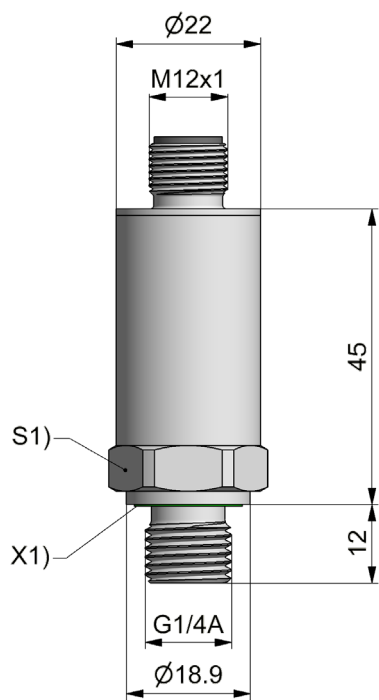
IO-Link Interface

Revision	IO-Link V1.1 Process Data Variable; Device Identification; Device Diagnosis
Min. process cycle time	4 ms
Transmission type	COM2, 38.4kBaud
Profile	Smart Sensor Profile 2 nd Edition v1.1.2
SIO-Mode	yes
Master port type	A
Process data analogue (in Pa)	2 Byte Process data 1 Byte scaling factor
Process data binary	1 byte
SDCI Standard	IEC 61131-9
Vendor ID	271 / 10f (hex)
Device IODD	https://ioddfinder.io-link.com/#/

M12x1

Protection class (mounted connector)		IP67
	Pin 1	V ₍₊₎
	Pin 2	S2 out
	Pin 3	0V / GND
	Pin 4	S1 out / IO-Link

SCPSi pressure switch



Order code

SCPSi Pressure switch

SCPSi-xxx-04-07

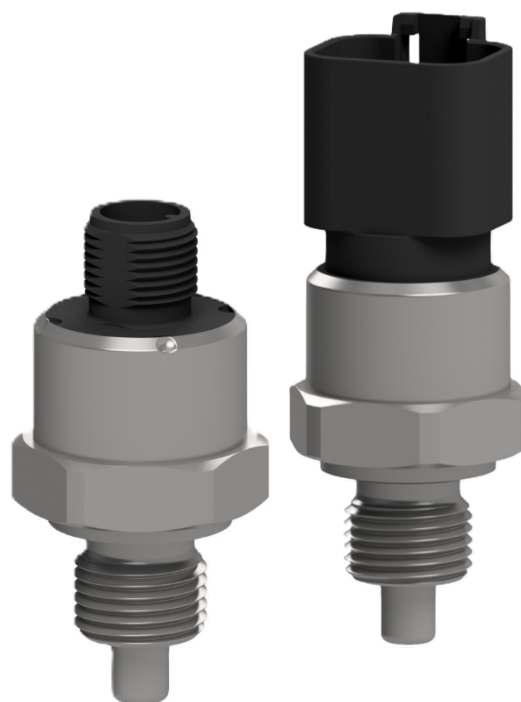
Druckbereich

0...001 bar	001
0...004 bar	004
0...010 bar	010
0...025 bar	025
0...060 bar	060
0...100 bar	100
0...250 bar	250
0...400 bar	400
0...600 bar	600

SCT03 temperature sensor

Device features

- Temperature sensor
- Wide temperature range
-50 ... +150°C
- Easy integration
 - Electrical outputs
 - Process connections
 - Electrical connectors
- High pressure resistance
 - Design
 - Robust
 - Compact
- Broad media compatibility
 - Stainless steel process connection
 - Liquids & gases



The SCT03 temperature sensor is engineered for the thermal and mechanical demands of mobile and industrial applications. It covers media temperatures from -50 to +150 °C and ambient conditions from -40 to +125 °C. With the high system pressure capability and IP67 ingress protection (mated), it is ready for harsh environments.

Industry-standard output signals and common connector/port options simplify integration.

Stainless-steel process connections enable broad media compatibility with liquids and gases.

The compact, robust metal housing makes the SCT03 well-suited for demanding applications such as mobile hydraulics.

Application examples

- Industrial hydraulics
- Mobile hydraulics
- Thermal management

SCT03 temperature sensor

Technical data

SCT03	007
Temperature range	-50 ... + 150°C
Max Pressure Pmax	600 bar
Response T05 / T90 [s]	< 10 / < 30
Material Housing	EN/DIN 1.4404; 316L
Weight	Approx. 45 g
RoHs	Yes
MTTFd	>100 years

Ambient conditions	
Media temperature	-50 to 150 °C
Operation temperature	-40 to 125 °C
Storage temperature	-40 to 125 °C
Vibration resistance	IEC 60068-2-6 : 20 g / 3 axes
Shock resistance	IEC 60068-2-31 (1m free fall)
CE Emission	EN 61326-1:2013-section 7 EN 61326-2-3:2013
CE susceptibility	EN 61326-1:2013 - section 6 EN 61326-2-3:2013

Process connection	Torque	Seal	Wetted parts	Pressure Range
G1/4A BSPP; DIN 3852 T11, Form E	25 Nm	FKM (green) *1 *2	316L (EN/DIN 1.4404) / FKM*1 *2	700bar
M10x1	25 Nm			350bar
9/16"-18 UNF, SAE 6	35 Nm			1000bar

*1 standard sealing FKM

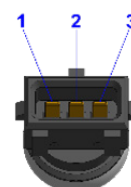
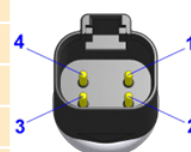
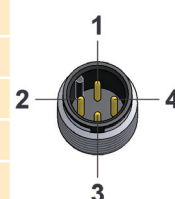
*2 EPDM for SCT03-XXX-XX-XX-EPDM instead of FKM

SCT03 temperature sensor

Electrical Connection

Output signal	4....20 mA	0,5...4,5 V ratio.	Pt100	Pt1000
Supply Voltage V+	10...32 VDC	5 V ±10%	-	-
Element	Pt100, class B	Pt100, class B	Pt100, class B	Pt1000, class A
Accuracy	@25°C ≤1% FS	@25°C ≤1% FS	DIN EN 60751 ± 0,3°C ± 0,005 x ΔT _{0-100°C}	DIN EN 60751 ±0,15°C±0,002 x ΔT _{0-100°C}
Accuracy @FS	≤2% FS	≤2% FS		
Load _{max}	< (V _{cc} -10 V)/20 mA	>4,7 kΩ		
Current consumption	≤22mA	max. 10 mA		
Reverse polarity	yes	yes		
Overvoltage protection	≤ 34V	≤ 32V		
M12 x 1 4 pole				
Pin 1	V+	V+	1/2 R+	1/2 R+
Pin 2	P-Signal	P-Signal	1/2 R+	1/2 R+
Pin 3	n.c.	0V / GND	3/4 R-	3/4 R-
Pin 4	n.c.	n.c.	3/4 R-	3/4 R-
Protection Class *	IP 67			
DT04-4P				
Pin 1	V+	V+	1/2 R+	1/2 R+
Pin 2	P-Signal	0V / GND	1/2 R+	1/2 R+
Pin 3	n.c.	P-Signal	3/4 R-	3/4 R-
Pin 4	n.c.	n.c.	3/4 R-	3/4 R-
Protection Class *	IP 67			
AMP JuniorTimer				
Pin 1	P-Signal	0V / GND	-	-
Pin 2	n.c.	P-Signal	-	-
Pin 3	V+	V+	-	-
Protection Class *	IP 67			

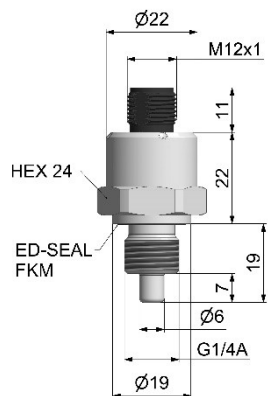
*connected/when plugged



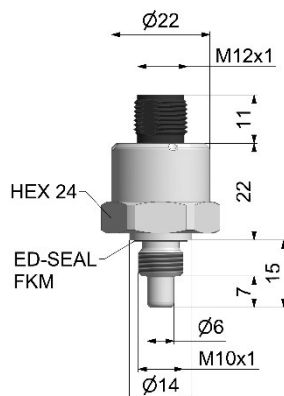
SCT03 temperature sensor

Drawings

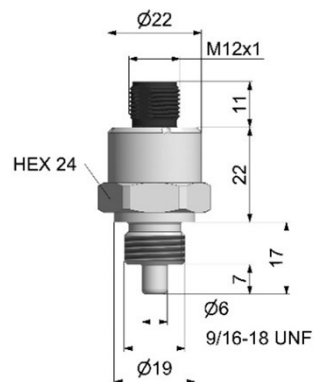
SCT03-007-X4-07



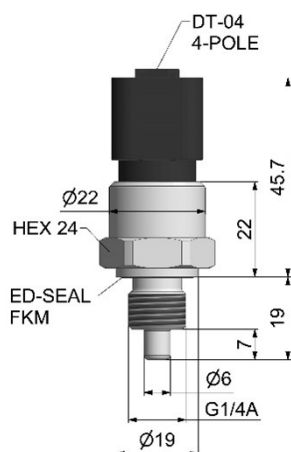
SCT03-007-XJ-07



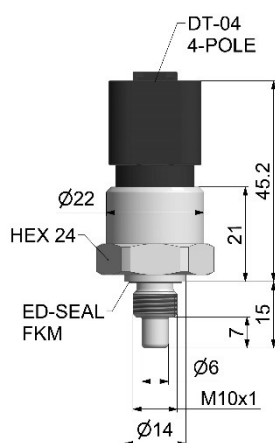
SCT03-007-X6-07



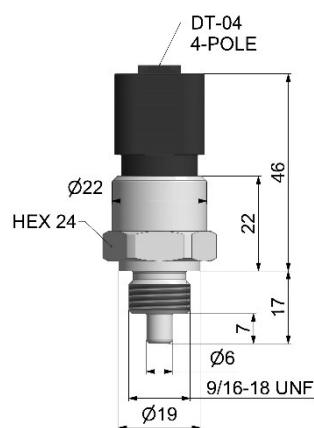
SCT03-007-X4-0D



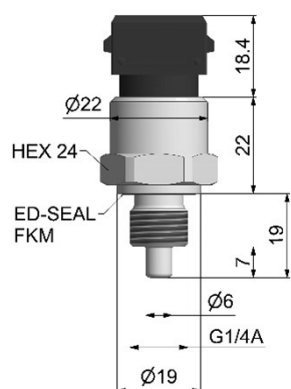
SCT03-007-XJ-0D



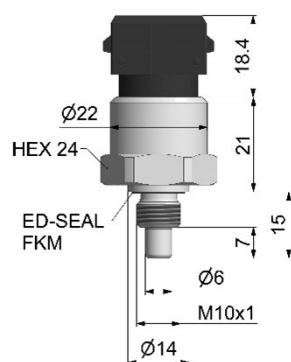
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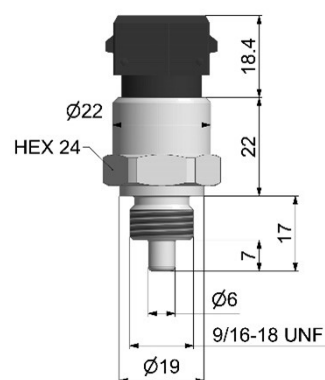
SCT03-007-X4-0J



SCT03-007-XJ-0J



SCT03-007-X6-0J



SCT03 temperature sensor

Order code

Temperature Sensor SCT03	SCT03 -007-	xx	0	x	-y
Output signal					
4...20 mA (2-wire)			3		
0.5...4.5 V (ratiometric)			R		
Pt100			H		
Pt100			T		
Process connection					
G1/4 BSPP			4		
M10x1			J		
9/16"-18 UNF, SAE 6			6		
Connecting plug					
M12x1 4-pole metallic			7		
DT04 4pole			D		
AMP Junior Timer 3-pole			J		
Sealing					
Standard FKM					
For EPDM extend part# with -EPDM					EPDM

*1 Only in combination with 9/16-18UNF-6M / 59°

Volumetric flow rate sensors

Device features

- Different measurement techniques
 - Quick
 - Not dependent on viscosity
 - Without loss
- Many measurement ranges
- Analogue output signal
- M12 connecting plug
- 24 VDC



The flow sensors used in **SensoControl®** provide accurate volume flow information in hydraulic systems (e.g. in testing equipment).

The sensors deliver a output signal that is proportional to the volumetric flow rate for further processing to an electronic system. They are compatible with conventional, well-known standards.

- M12 connecting plug
- 24 VDC
- 0/4 to 20 mA

The volumetric flow rate can be easily displayed when using the **SCE-020** panel meter.

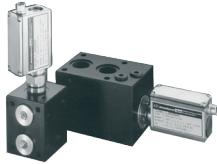
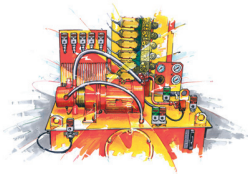
In order to meet the many different application requirements, three different measuring principles are available:

- **SCVF** geared counter
- **SCFT** turbine
- **SCQ** spring/piston

The volumetric flow rate sensors are used in control, regulation or monitoring systems where analogue signals are needed to capture the volume flow.

Volumetric flow rate sensors

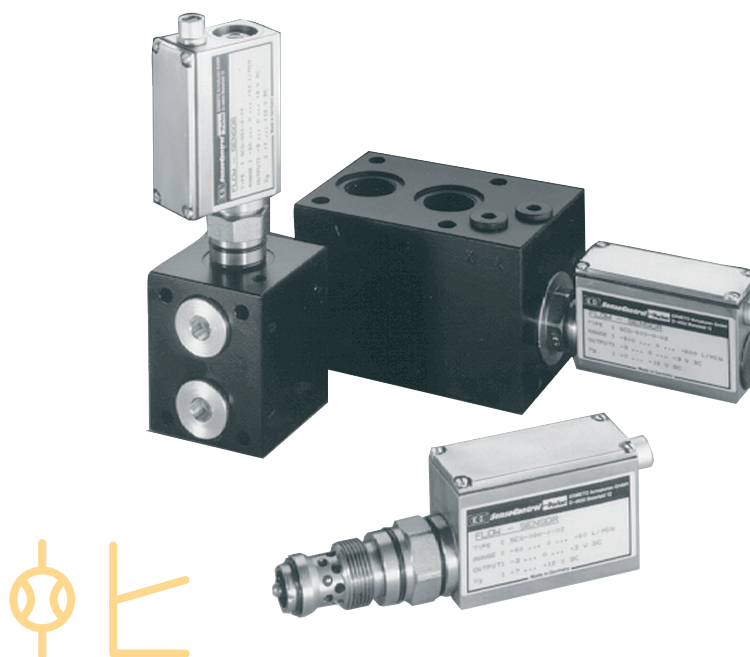
Overview

	SCQ	SCFT	SCVF
			
Range of use	For quick flow changes Measures in both directions - Response speed $\leq 2\text{ ms}$ - Reverse operation - Wide viscosity range - Compact size - Up to 420 bar (6092 psi)	Low loss measuring of volume flow - Response speed $\leq 50\text{ ms}$ - Many measurement ranges - Low flow resistance - Up to 800 l/min - Up to 420 bar (6092 psi)	Measures different substances Measures lower volume flows (leakage measurements) - Very wide measurement range - Not dependent on viscosity - Up to 400 bar (5802 psi)
Applications	- Test rigs - General machine construction - Hydraulic plant construction 		
Order code	SCQ-xxx-10-07	SCFT-xxx-22-07	SCVF-xxx-10-07
Refer to page	46-49	50-53	54-59

SCQ flow meter

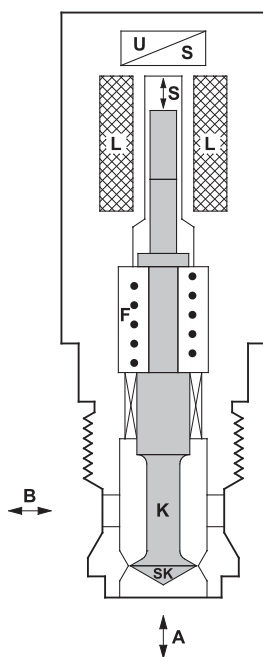
Device features

- Measurement principle Spring/piston principle
- Response time ≤ 2 ms
- Measurement in both directions
- Wide viscosity range
- Compact design
- Withstands pressures up to 420 bar (6092 psi)



Function

The piston (K) is moved due to a flow from A to B or from B to A. In the idle state, the spring (F) and the piston (K) are in equilibrium. The delta (S) is proportional to the flow and is converted to a value through the built-in electronics. Through the change in direction of the piston (B to A), the flow direction can be indicated. (e.g. -45.8 l/min) The reaction time of the piston movement is less than 2 ms.



SCQ measurement principle

Application

When working with high-pressure hydraulics, it is very important to be able to quickly detect the flow rate.

Installation with a connection block permits the combined measurement of p, T and Q. Rapid assembly of the **SCQs** is achieved with an in-line adaptor for tube or hose installation. Use under extreme conditions (such as high load changes or rapid pressure increases) is possible because of the sturdy construction.

The **SCQ** is the perfect solution when recording highly dynamic volume flow changes. Rapid load changes, which can cause damage for example in valves and pumps, can be safely detected. Due to its unique measurement process, the **SCQ** can capture volume flow in both directions.

SCQ flow meter

Technical data

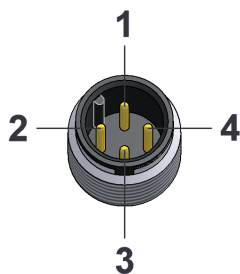
SCQ-	150
Measuring range QN	-150...+150 l/min
Q _{max}	-165...+165 l/min
Substance connection	M42 (NG16)
Weight (g)	1050

Accuracy	
Deviation from characteristic curve	±2 % FS @ 46cSt.
Response time	2 ms
Thermal drift	±0.05 % FS/°C
Repeat accuracy	± 0.5 % FS
Resistance to pressure	
Pressure range	3...420 bar
Operating pressure P _n	315 bar / (4569 psi)
Overload pressure P _{max}	420 bar / (6092 psi)
Pressure drop ΔP (bar) @ (FS)	Refer to diagram
Material	
Housing	Steel
Seal	NBR
Parts in contact with substances	Steel, NBR
Ambient conditions	
Operating temperature	+10...+60 °C / (50...140°F)
Storage temperature	-20...80 °C / (-4...176°F)
T _{max} Fluid	+80 °C / (176°F)
Filtration	25 μm

Viscosity range	15...100 cSt.
Protection degree	IP67 DIN EN 60529
Electrical connection	
Plug	M12x1; 4-pole
Supply voltage	+18...+30 VDC
Current consumption	40 mA
Output	0...20 mA = -FS...+FS (10 mA = 0 l/min)
Load	≤ 150 Ω
Signal noise	< 5 mV
EM compatibility	
Disturbance emissions	EN 61000-6-3
Resistance to interference	EN 61000-6-2

Pin assignment

M12x1; 4-pole

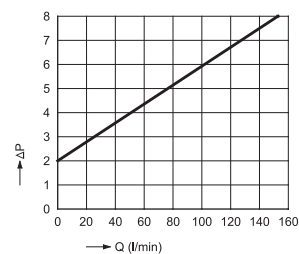
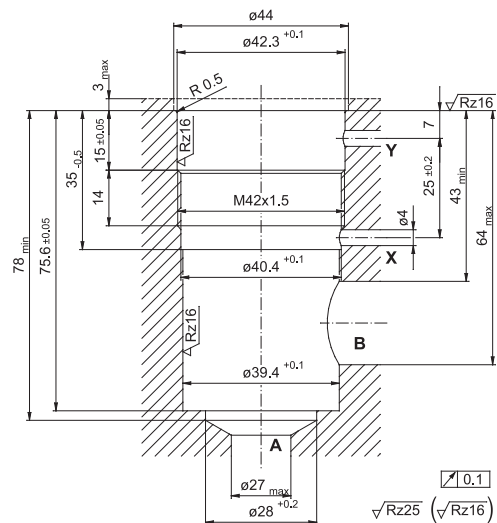
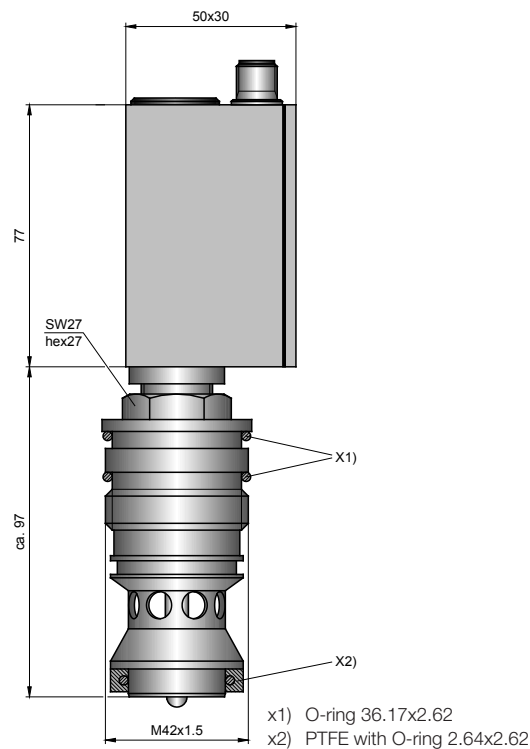


PIN	Assignment
1	V ₊
2	Q signal
3	0 V/GND
4	—

SCQ flow meter

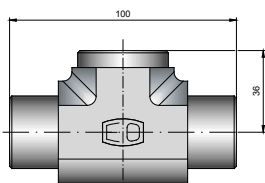
Screw plug hole and pressure-drop curve **SCQ-150**

30 Nm torque

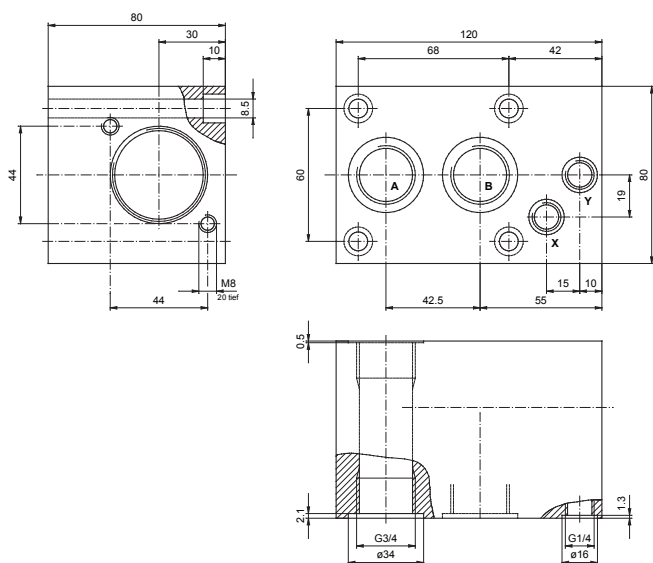


SCQ flow meter

SCAQ-GI-R1/2



SCAQ-150



Order code

SCQ-150 (-150 to +150 l/min)

M12x1, 4-pole; connecting plug; IP67
0 to 20 mA; -150...+150 l/min

SCQ-150-10-07

Accessories SCQ-150

Connector block
G3/4 BSPP inner (A-B) and M42 inner
With screw plug:
M42 outer and
G3/4 BSPP outer (A-B)

SCAQ-150

Spare parts

Spacer ring for SCQ-060
Seal kit for SCQ-060
Seal kit for SCQ-150

SC-910

SC-911

SC-912

Connection cable and single plug

Connection cable, assembled

(open cable end)

SCK-400-xx-xx

Cable length (m)

2 m	02
5 m	05
10 m	10

Connecting plug

M12 cable jack; straight	45
M12 cable jack; 90° angled	55

Single connector

M12 cable jack; straight
M12 cable jack; 90° angled

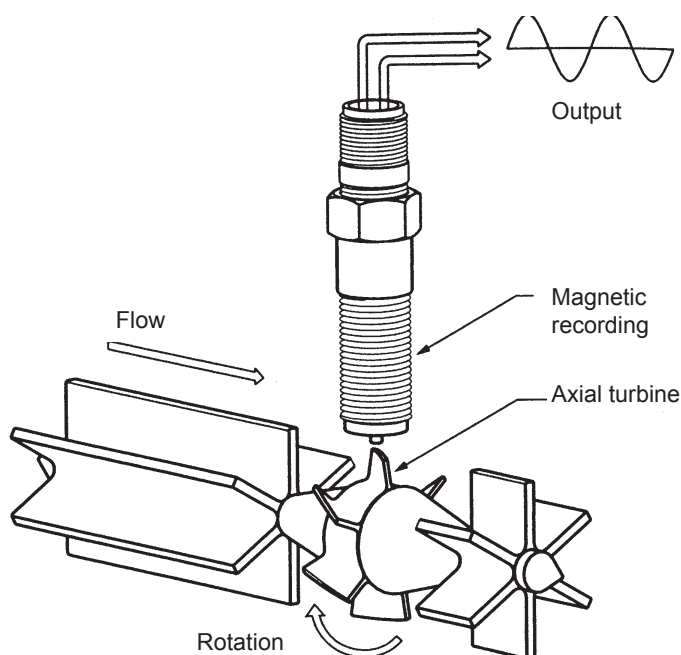
SCK-145

SCK-155

SCFT measurement turbine

Device features

- Measurement principle: Turbine
- Response speed ≤ 50 ms
- Measurement range from 1 to 800 l/min
- Low flow resistance
- Suitable for reverse operation
- Built-in pressure and temperature ports



Function

The turbine wheel is driven by the oil flow. The generated frequencies are processed through the digital electronics and influences from the disturbing flow effects are compensated for. Because of the low flow resistance Q_R , the hydraulic circuit operates with very low losses.

Reverse operation is also possible because of the special vane (winged) design - so the turbine can be operated in both directions.

The turbine is fitted with an EMA-3 screw coupling for measuring pressure. Oil temperature can be measured directly in the oil flow of the turbine by connecting the temperature sensor (**SCT-150**). This provides all important measurements at the installation location.

Application

The **SCFT** is the ideal solution if the volumetric flow rate needs to be recorded loss-free across a wide flow range (up to 800 l/min.).

SCFT measurement turbine

Technical data

SCFT-	015	060	150	300	600	800
Flow measuring range Q _n (l/min) (US Gal/min)	1...15 (0.25...4)	3...60 (0.8...16)	5...150 (1.3...40)	8...300 (2...80)	15...600 (4...160)	20...800 (5...210)
Accuracy (± %) FS/IR @ 21cSt.	± 1 % FS	± 1 % IR	± 1 % IR	± 1 % IR	± 1 % IR	± 1 % IR
Operating pressure P _n bar / (psi)	350 (5076)	350 (5076)	350 (5076)	350 (5076)	290 (4206)	400 (5801)
Ports (A - B)	G1/2 BSPP	G3/4 BSPP	G3/4 BSPP	G1 BSPP	G1 1/4 BSPP	G1 7/8 UNF
Pressure drop ΔP (bar) @ (FS)	1.5	1.5	1.5	4	4	5
Weight (g)	700	1600	1600	1700	2700	5000

FS = Full Scale

IR = Indicated Reading

Accuracy

Response time	50 ms
Thermal drift	±0.05 % FS/°C
Repeat accuracy	± 0.5 % FS

Resistance to pressure

Q _{max} (l/min)	Q _N x 1.1
Overload pressure P _{max}	P _N x 1.2

Material

Housing	Aluminium
Seal	FKM
Parts in contact with sub- stances	Aluminium, steel, FKM

Ambient conditions

Ambient temperature	-10...+50 °C / (14...122°F)
Storage temperature	-20...+80 °C / (-4...176°F)
T _{max} Fluid	-20...+80 °C / (-4...176°F)
Filtration	25 µm (10 µm for SCFT-015)
Viscosity range	15...100 cSt.
Protection class	IP66 EN60529

Ports

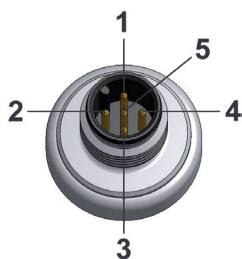
Temperature measurement (SCT-150-14-07)	M10x1 OR
Pressure connection	EMA3
Pressure (VSTI)	G1/4 BSPP

Electrical connection

Plug	M12x1; 5-pole
Power supply V ₊	18...30 V
Output signal	4...20 mA ± 0...FS l/min
Complete output current range	0...21 mA
Current consumption	< 30 mA
Protection degree	IP66 EN60529

Pin assignment

M12x1; 5-pole

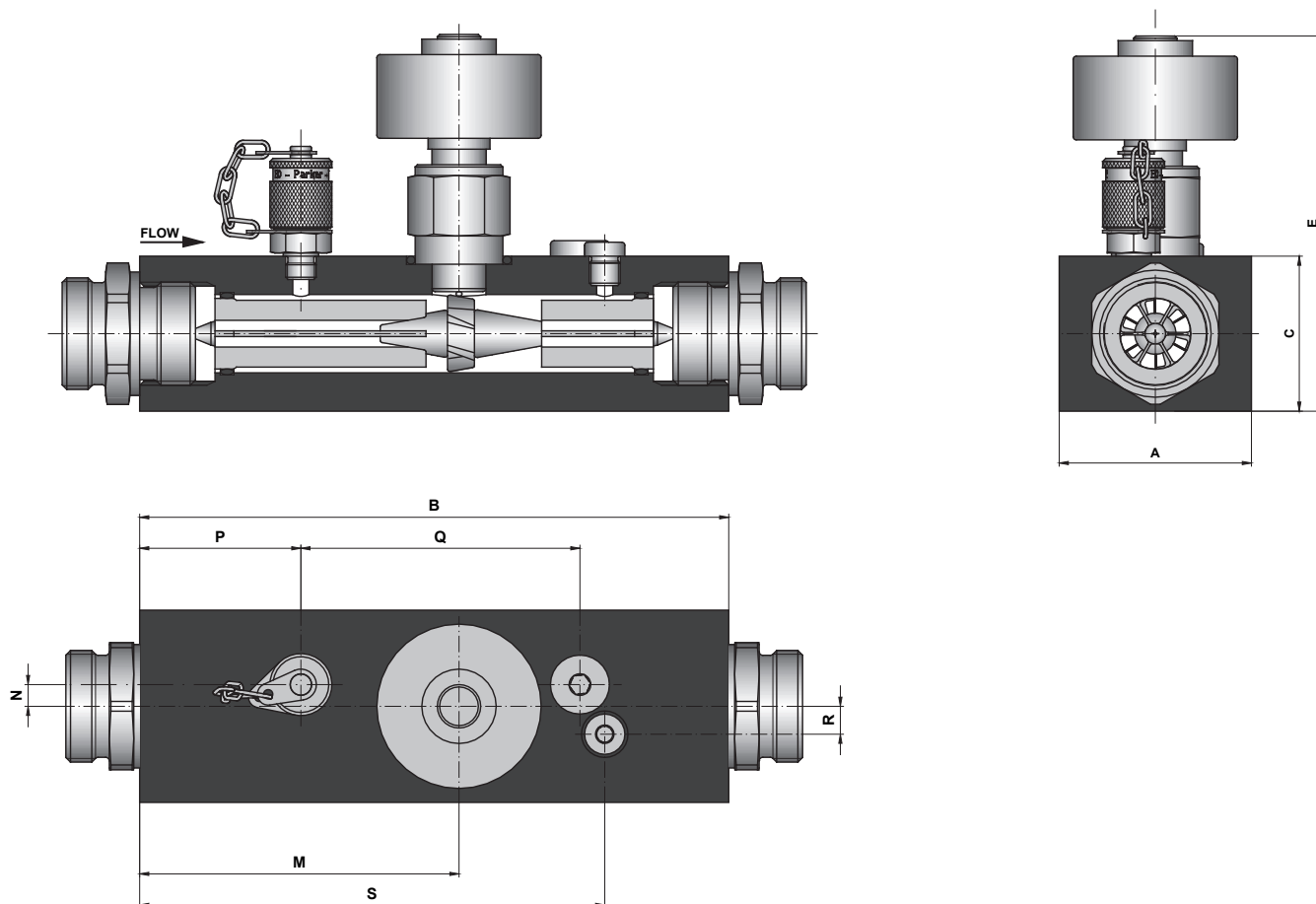


PIN	Assignment
1	V ₊
2	n.c.
3	Q signal
4	n.c.*
5	0 V / GND

*n.c. = do not connect



SCFT measurement turbine



#	SCFT-015	SCFT-060	SCFT-150	SCFT-300	SCFT-600	SCFT-800
A	37	62	62	62	62	100
B	136	190	190	190	212	212
C	37	50	50	50	75	75
E	115	130	130	134	149	152
M	70	103	103	103	127	126
N	0	5	5	7	9	10
P	25	50	50	52	62	60
Q	N/A	92	92	90	106	104
R	0	5	5	9	11	10
S	115	157	157	150	168	181

SCFT measurement turbine

Order code

SCFT

M12x1, 5-pole; connecting plug; IP66

4...20 mA (3-wire)

BSPP:

1...15 l/min	SCFT-015-22-07
3...60 l/min	SCFT-060-22-07
5...150 l/min	SCFT-150-22-07
8...300 l/min	SCFT-300-22-07
15...600 l/min	SCFT-600-22-07
20...800 l/min	SCFT-800-22-07

UNF:

0.25...4 US Gal/min	SCFT-004G-2U-07
0.8...16 US Gal/min	SCFT-016G-2U-07
1.3...40 US Gal/min	SCFT-150G-2U-07
2...80 US Gal/min	SCFT-080G-2U-07
4...160 US Gal/min	SCFT-160G-2U-07
5...210 US Gal/min	SCFT-210G-2U-07

Connection cable and single plug

Connection cable, assembled

(open cable end)

SCK-400-xx-xx

Cable length (m)

2 m	02
5 m	05
10 m	10

Connecting plug

M12 cable jack; straight	45
M12 cable jack; 90° angled	55

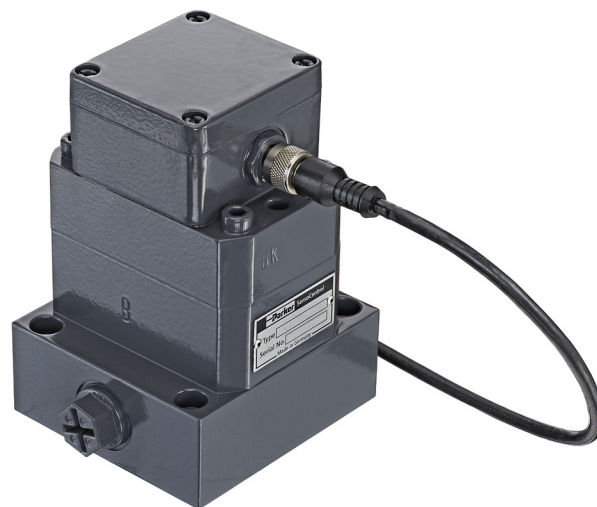
Single connector

M12 cable jack; straight	SCK-145
M12 cable jack; 90° angled	SCK-155

SCVF volume counter

Device features

- Measurement principle: Volume/geared counter
- Eight measurement ranges from 0.01 - 2 to 1 - 300 l/min
- Accuracy $\pm 0.5\%$ FS
- Withstands pressures up to 400 bar (5802 psi)
- High viscosity range
- Low noise
- Exact flow rate measurement over a wide viscosity range
- Versatile usage for different substances



Gear counter for highly accurate flow rate measurements in hydraulic systems

Function

The SCVF geared counter functions as a volume flow meter. A very precisely crafted pair of geared wheels is driven by the fluid flow.

The SCVF works over a wide viscosity range. Different seals permit usage in many different applications.

Applications

Due to the wide viscosity range, any liquid can be measured that can be pumped and has a certain degree of lubricating capability.

- Brake fluid (EPDM seal)
- Skydrol
- Mineral oils
- Hydraulic oil and
- Grease

The SCVF is the ideal solution when carrying out precise flow rate measurements over a wide viscosity range.

SCVF volume counter

Technical data

SCVF-	002	004	015	040	060	080	150	300
Flow measuring range (l/min)	0.01...2.0	0.02...4.0	0.2...15	0.4...40	0.4...60	0.4...80	0.6...150	1.0...300
Pressure range P _N bar / (psi)	400 (5802)	315 (4569)	400 (5802)	400 (5802)	400 (5802)	400 (5802)	315 (4569)	315 (4569)
Overload pressure P _O bar / (psi)	480 (6962)	400 (5802)	480 (6962)	480 (6962)	480 (6962)	480 (6962)	350 (5076)	350 (5076)
Connection	G3/8 BSPP	G3/8 BSPP	G3/8 BSPP	G1/2 BSPP	G1/2 BSPP	G1/2 BSPP	G1 BSPP	G1 BSPP
Sound level dB (A)	< 60	< 60	< 60	< 70	< 70	< 70	< 70	< 72
Resolution (pulses / litre)	40,000	25,000	4082	965	965	965	333.33	191

Accuracy

Deviation from characteristic curve	± 0.3 % FS ≥ 20 cSt. ± 0.5 % FS ≥ 20 cSt.
Response time	< 10 ms
Repeat accuracy	0.01 % FS
Substance *)	Hydraulic oil (25 micron filter)

Material

	Material 1.7139 Contains no non-ferrous metal or silicone
Housing	Steel
Seal	FKM EPDM on request

Ambient conditions

Ambient temperature	0...+55 °C / (32...131°F)
Storage temperature	-25...+85 °C / (-13...185°F)
Fluid temperature	-30...120 °C / (-22...148°F)
Viscosity range	Refer to diagram p. 44
Protection degree	IP65 DIN EN 60529

FS = Full scale value

*) When using other substances, please state the viscosity range and the type of seals. (Attach the data sheet of the substance if possible)

Electrical connection

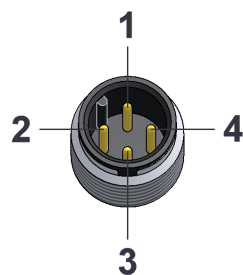
Plug	M12x1; 4-pole
Power supply V ₊	+18...+30 VDC
Current consumption	< 28 mA
Output signal	0...20 mA ± 0...FS l/min
Load	≤ 150 Ω

EM compatibility

Disturbance emissions	EN 61000-6-3
Resistance to interference	EN 61000-6-2

Pin assignment

M12x1; 4-pole

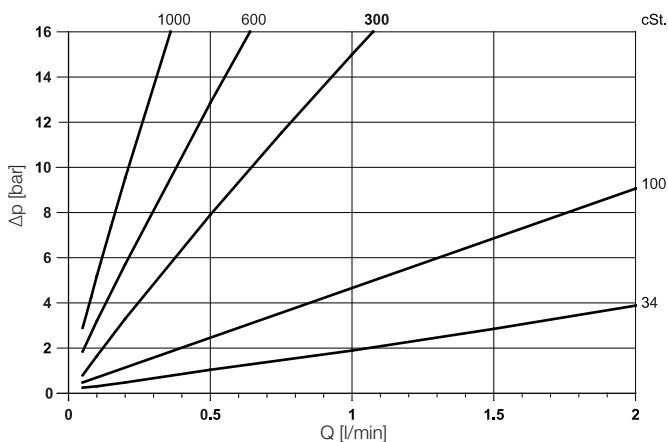


PIN	Assignment
1	V ₊
2	Q-signal
3	0 V / GND
4	—

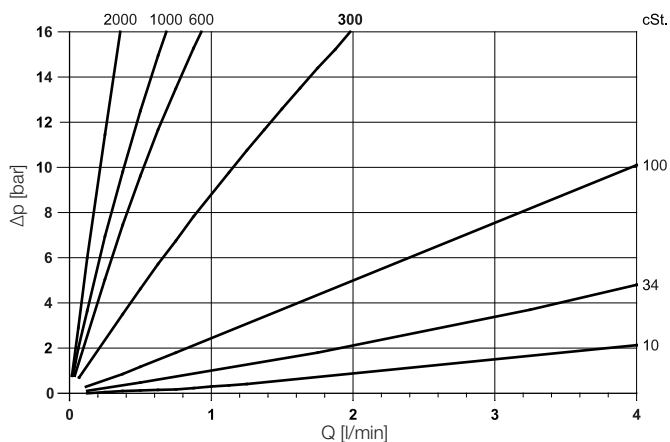
SCVF volume counter

Technical data

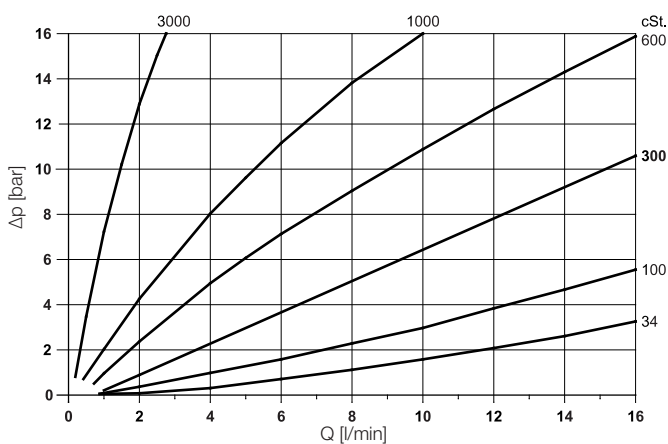
SCVF-002 Δp - Viscosity



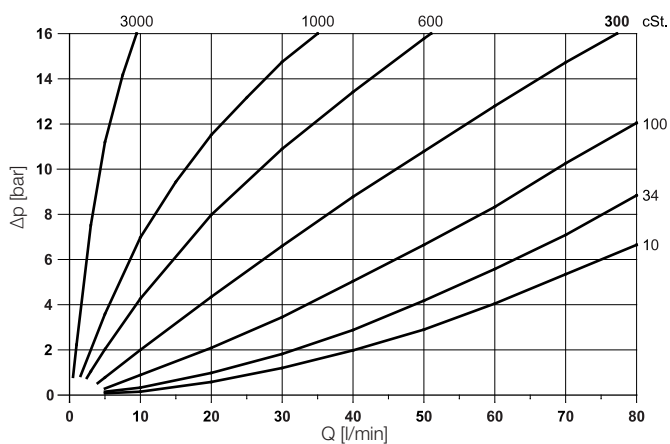
SCVF-004 Δp - Viscosity



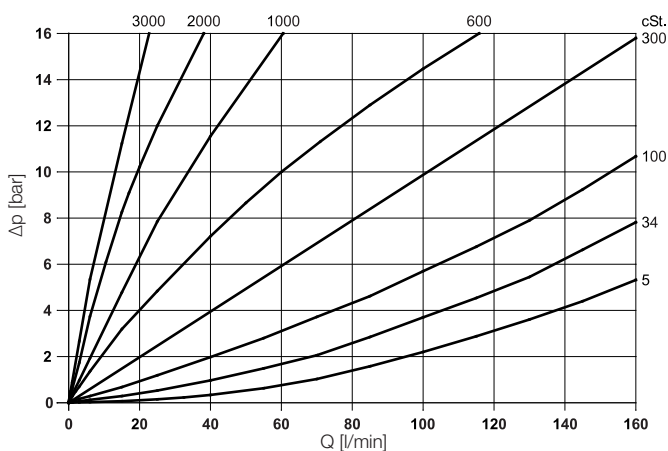
SCVF-015 Δp - Viscosity



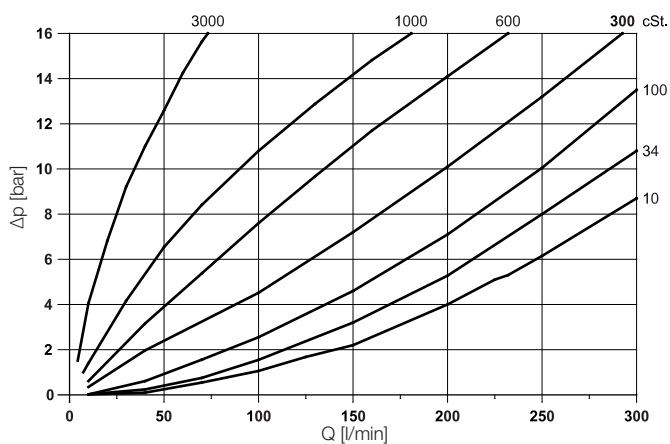
SCVF-040/060/080 Δp - Viscosity



SCVF-150 Δp - Viscosity



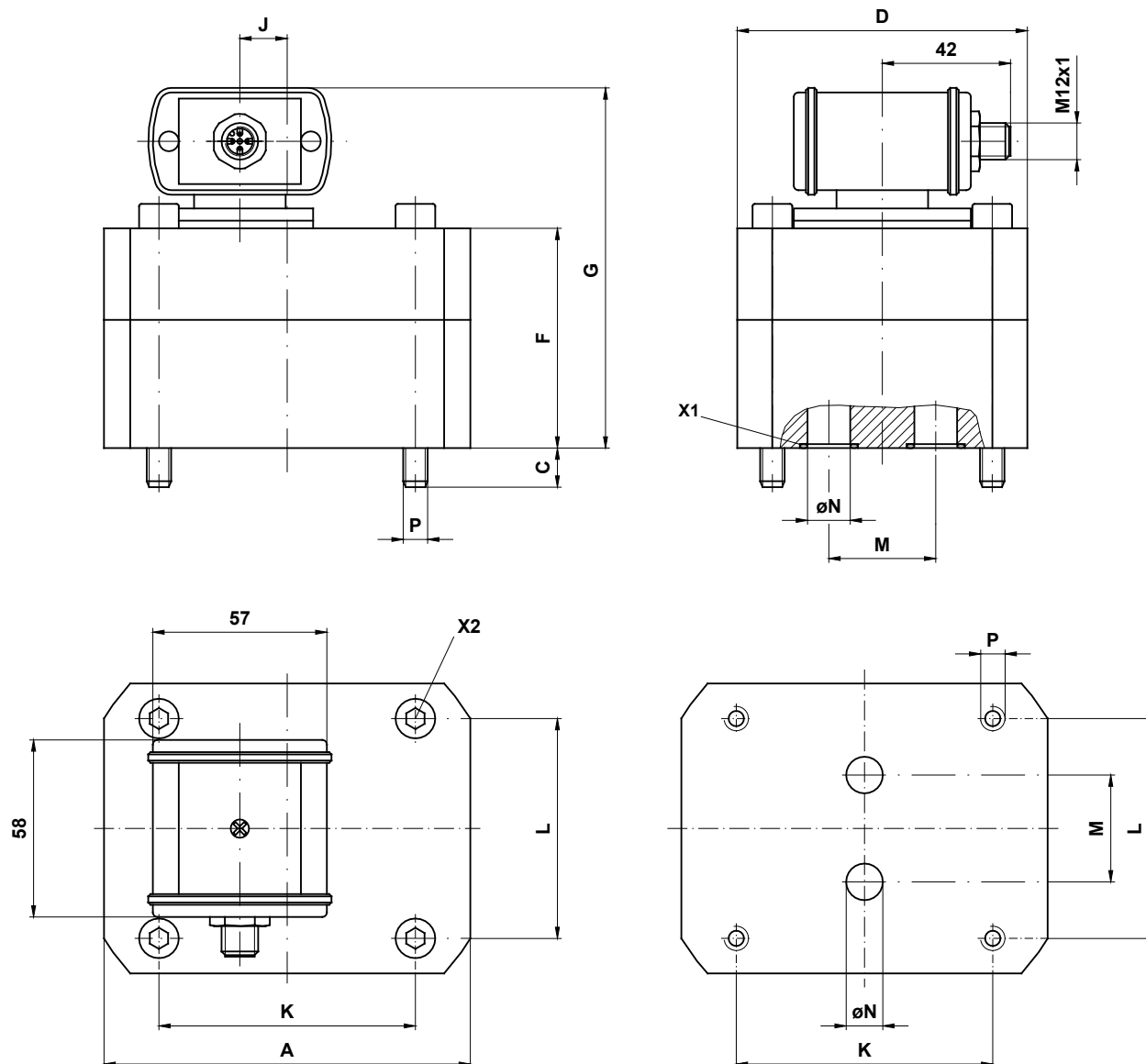
SCVF-300 Δp - Viscosity



Δp = pressure loss



SCVF volume counter

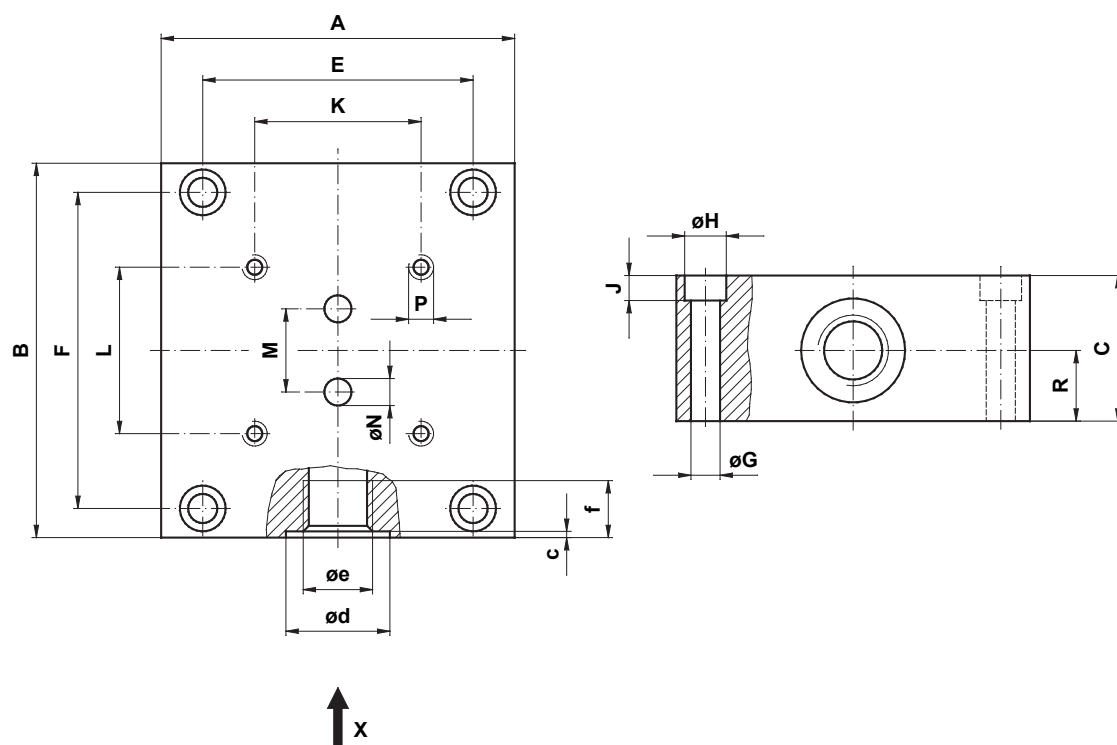


Type	Weight [kg]	Torque [Nm]	A	C	D	F	G	J	K	L	M	øN	P
SCVF-002	1.8	14	85	10	60	50	87	-	70	40	20	6.5	M6
SCVF-004	2	14	85	9	60	56		-	70	40	20	6.5	M6
SCVF-015	2	14	85	13	60	57	94	-	70	40	20	9	M6
SCVF-040	5.2	35	120	13	95	72	109	10.5	84	72	35	16	M8
SCVF-060													
SCVF-080													
SCVF-150	9	120	170	18	120	89	140	46.5	46	95	50	25	M12
SCVF-300	13	120	170	22	120	105	142	40	46	95	50	25	M12

All measurements in mm

SCVF volume counter

Dimensioned drawings connection plate



Type	kg	A	B	C	E	F	ØG	ØH	J	K	L	M	ØN	P	R	c	Ød	Øe BSPP	f
SCVF-002 SCVF-004 SCVF-015	1.8	85	90	35	65	76	7	11	7	70	40	20	6.5	M6/t = 14	17	0.7	25	G3/8	13
SCVF-040 SCVF-060 SCVF-080	2.9	100	120	37	80	106	7	11	7	84	72	35	12	M8/t = 18	17.5	0.7	29	G 1/2	15
SCVF-150 SCVF-300	14	160	165	80	140	145	9	15	9	46	95	50	25	M12/t = 24	28	1	42	G1	19

All measurements in mm

SCVF volume counter

Order code

SCVF

M12x1, 4-pole; connecting plug; IP65; incl. connection plate

0...20 mA

0.01...2 l/min

SCVF-002-10-07

0.02...4 l/min

SCVF-004-10-07

0.2...15 l/min

SCVF-015-10-07

0.4...40 l/min

SCVF-040-10-07

0.4...60 l/min

SCVF-060-10-07

0.4...80 l/min

SCVF-080-10-07

0.6...150 l/min

SCVF-150-10-07

1...300 l/min

SCVF-300-10-07

Connection cable and single plug

Connection cable, assembled

SCK-400-xx-xx

(open cable end)

Cable length (m)

2 m

02

5 m

05

10 m

10

Connecting plug

M12 cable jack; straight

45

M12 cable jack; 90° angled

55

Single connector

M12 cable jack; straight

SCK-145

M12 cable jack; 90° angled

SCK-155

I/O-Link Switch Family

Device features

- Large display
- Freely adjustable
- Rugged metal construction
- Compact size
- Long-term stability
- Dependable
- Immune to interference



This switch is used in control, regulation or monitoring systems where switching signals or analogue signals are used or a display is required.

The switch can replace the following:

- Mechanical switches
- Mechanical displays
(pressure gauges, thermometers, inspection glass)
- Sensors

All the above mentioned functions can be combined in one device.

All switch devices have a compact and pivoting metal housing so that they can be mounted optimally under adverse installation conditions. The large display can always be perfectly positioned so that it is easy to read even at longer distances.

Both of the switching outputs can be set individually either as NO or NC. They also both have hysteresis and the window functions. Therefore the on and off switching values as well as delay times (attenuation) for each of the switching points can be chosen freely.

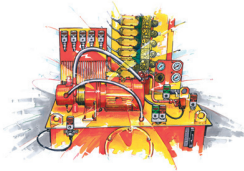
Thanks to these easy switching functions, intelligent adjustments can be set which are normally not possible using a mechanical switch. Therefore, many switches can be replaced with one product of the I/O-Link switch family.

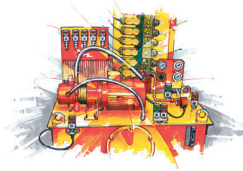
The switches offer good practical characteristics combined with diverse mounting and setting options.

Because of its compact design, long lifespan and high functionality, this controller is ideal for the permanent series use in hydraulic and pneumatic applications.

I/O-Link Switch Family

Overview

	SCPSDI	SCTSDI (Modular)
		
Range of use	Pressure display and monitoring ■ Compact size ■ Resistant to pressure peaks ■ Resistant to shock and vibration ■ IO-Link interface	Temperature display and monitoring ■ Temperature display ■ Fixed level contacts
Applications	■ Test benches ■ Processing equipment ■ Conveying and lifting equipment ■ General machine construction ■ Pneumatic plant construction ■ Hydraulic plant construction 	
Order code	SCPSDI-xxx-x4-xx	SCTSDI-150-1T-05
Refer to page	62-65	70-74

	SCLSDI	SCLTSDI	SCTSDI
			
Range of use	Level indication and monitoring ■ Level display ■ Practical monitoring with window function ■ Continuous level measurement	Level/temperature display and monitoring ■ Level display ■ Temperature display ■ Continuous level measurement ■ One bore hole	Temperature display and monitoring ■ Temperature display ■ Practical monitoring with window function ■ Continuous temperature measurement
Applications	■ Test benches ■ Processing equipment ■ Conveying and lifting equipment ■ General machine construction ■ Pneumatic plant construction ■ Hydraulic plant construction 		
Order code	SCLSDI-xxx-x0-xx	SCLTSDI-xxx-x0-xx	SCTSDI-xxx-x0-xx
Refer to page	75-78	79-82	66-69

SCPSDI Pressure Switch

Device features

Display

- Large Display
- 2-color switching status display
- Reversal of viewing direction by 180°
- Beveled for better readability
- Units in bar, MPa and psi

Easy to use

- 3 tactile buttons
- VDMA-Menu navigation

Outputsignal

- IO-Link interface
- Stable metallic M12x1 thread
- Pin assignment according to VDMA
- Can be selected between switching output, current output and voltage output

Measuring element

- Pressure range up to 600 bar possible
- High compressive strength
- High accuracy up to 0,5% FS
- Long-term stable measuring cell

Thread

- G1/4 internal and external thread
- Stainless steel 17-4PH = /1.4542
- Broad media compatibility

Housing

- Metallic housing
- High quality surface
- 240° rotatable
- IP67 protection
- Increased temperature range between -40°C up to +85°C



SCPSDI Pressure Switch

Technical data

SCPSDI-	004	010	016	060	100	250	400	600
Pressure range P_n	-1...4	-1...10	-1...16	0...60	0...100	0...250	0...400	0...600
Overload pressure P_{max}	50	50	50	500	500	800	1200	1200
Burst pressure P_{burst}	1500	1500	1500	2500	2500	3000	3500	3500
Display resolution	0,001	0,01	0,01	0,01	0,1	0,1	0,1	0,1
Min. adjustable difference SP and rSP	0,01	0,01	0,01	0,1	0,1	0,1	1	1

All units in bar

Process connection

Process connection	G1/4 BSPP Internal/External
Tightening torque	35 Nm

Parts in contact with media

Internal thread	Stainless Steel: 17-4PH = / 1.4542
External thread	Stainless Steel: 17-4PH = / 1.4542
	Sealing: ED FKM SH75
Sampling rate	≤3 ms
Medium temperature	-40...+85 °C
Switching cycles	≥10 Mio.

Output variables

Current output	0..20mA or 4..20mA (programmable)
Voltage output	0..10V

Switching point

Switching current:	max. 200 mA
Short circuit current:	400mA (short term), short circuit proof
Switching voltage:	Supply voltage - 0,5 VDC
Execution:	PNP/ NPN (programmable)
Max switching frequency	250 Hz

Accuracy

@ 25°C	±0,5% FS
@ -20...+85°C	±1,5% FS

Display accuracy

@ 25°C	±0,5% FS ±1 Digit
@ -20...+85°C	±1,5% FS ±1 Digit

Response speed

Switching output	≤3 ms
Analogue output	≤3 ms

Electrical connection

Supply voltage V_+	18...30 VDC
Connector	M12x1; 4pol / M12x1; 5pol
Short circuit protection	Yes
Reverse polarity protection	Yes
Overload protection	Yes
Overvoltage	50 V
Current consumption	<40 mA @24V
Inrush current	<100 mA @24V

IO-Link

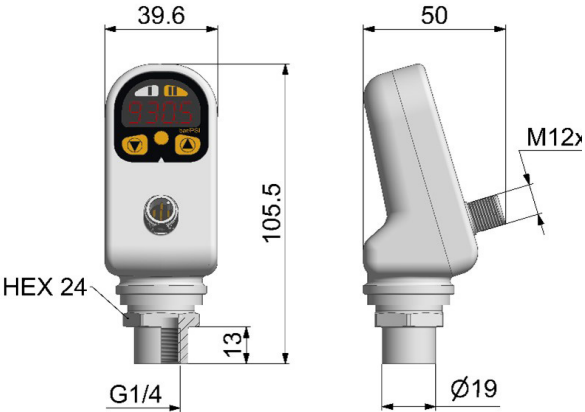
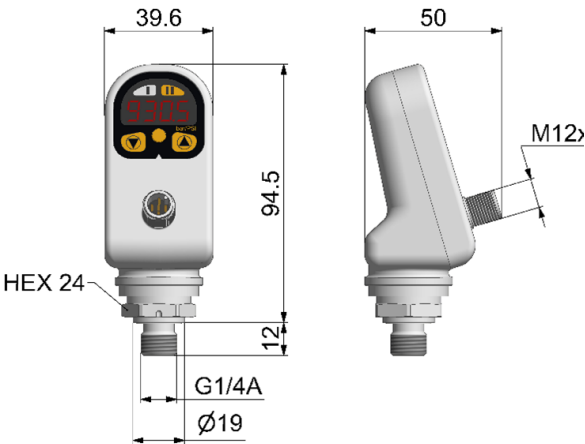
Profile:	Smart Sensor Profile 2nd Edition V1.1.2
Process Cyklustime	4 ms
SIO-Mode	Yes
SIO-Mod	A
Process data analogue	2 Byte Process data 1 Byte Scaling factor
Process data binary	1 Byte
SDCI Standard	IEC 61131-9
Vendor ID	271 (dec) / 10f (hex)
Device IODD	https://parker.com/

SCPSDI Pressure Switch

Technical data

Housing	
Rotatability	>240°
Readability display	Reversal of viewing direction by 180° (programmable)
Display	7-Segment-LED (4-digit)
	2-color switching status display (red/ yellow)
Buttons	Digit height : 8 mm
	3 tactile buttons for up, down, enter
Material	
Housing	Zinc die-cast (bright nickel-plated)
Display cover	Polyester Autotex V207
Protection class	IP67
Weight	150 g
Environmental conditions	
Ambient temperature	-40...+85 °C
Operating temperature	-40...+85 °C
Storage temperature	-40...+85 °C
Vibration resistance	DIN EN 60068-2-6, 20 g
Shock resistance	DIN EN 60068-2-27, 50 g

EM compatibility	
Immunity to interference	
Emission of interference	EN 55011
ESD	EN 61000-4-2
high-frequency electro-magnetic fields	EN 61000-4-3
fast transient electrical disturbances	EN 61000-4-4
Conducted disturbances induced by high-frequency fields	EN 61000-4-6
General	
MTTFd	>100 Jahre
RoHS-Conformity	2011/65/EU, 2024/232



SCPSDI Pressure Switch

Order code

Digital PressureSwitch SCPSDI

2 switching points; no analogue output SCPSDI-xxx - 04 - x7
M12x1 connector; 4-Pin

1 switching point; with current output SCPSDI-xxx - 14 - x7
M12x1 connector; 4-Pin

2 switching points; with current output or voltage output SCPSDI-xxx - x4 - x5
M12x1 connector; 5-Pin

Pressure range

004 bar	004
010 bar	010
016 bar	016
025 bar	025
060 bar	060
100 bar	100
250 bar	250
400 bar	400
600 bar	600

Analogue output

No analogue output	0
0..20 mA / 4..20 mA	1
0..10 V	4

Version

G1/4 BSPP External thread	1
G1/4 BSPP Internal thread	2

Accessory

Securing clamp
Reducing adapter M22x1,5
Reducing adapter G1/2 BSPP
Attenuation adapter

SCSD-S24
RI22X1.5EDX1/4CF
SCA-1/4-ED-1/2-ED
SCA-1/4EDX1/4-D

Order example

SCPSDI-100-04-27

Pressure range 100 bar
2 switching points
G1/4 BSPP Internal thread
M12 Connector 4-Pin

SCPSDI-004-44-15

Pressure range 4 bar
2 switching points
1 analogue output 0..10V
G1/4 BSPP External thread
M12 Connector 5-Pin

Connection cable and single plug

Connection cable, assembled
(open cable end)

SCK-400 - xx - xx

Cable length (m)

2 m	02
5 m	05
10 m	10

Connector

M12 cable jack; straight	45
12 cable jack; 90° angled	55

Single connector

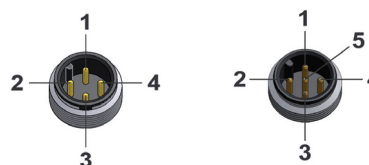
M12 cable jack; straight	SCK-145
M12 cable jack; 90° angled	SCK-155

Connection

PIN	Assignment SCPSDI-xxx-x4-x7
1	V ₊
2	Analogue out
3	0 V/GND
4	S1 out / IO-Link

PIN	Assignment SCPSDI-xxx-04-x7
1	V ₊
2	S2 out
3	0 V/GND
4	S1 out / IO-Link

PIN	Assignment SCPSDI-xxx-x4-x5
1	V ₊
2	S2 out
3	0 V/GND
4	S1 out / IO-Link
5	Analogue out



SCTSDI Temperature Switch

Device features

Display

- Large displays for temperature
- 2-color switching status display
- Reversal of viewing direction by 180°
- Bevelled for better readability
- Units in °C and °F

Easy to use

- 3 tactile buttons
- VDMA-Menu navigation

Outputsignal

- IO-Link interface
- Stable metallic M12x1 thread
- Pin assignment according to VDMA
- With 2 switching outputs and switchable current output between 0..20 and 4..20mA

Housing

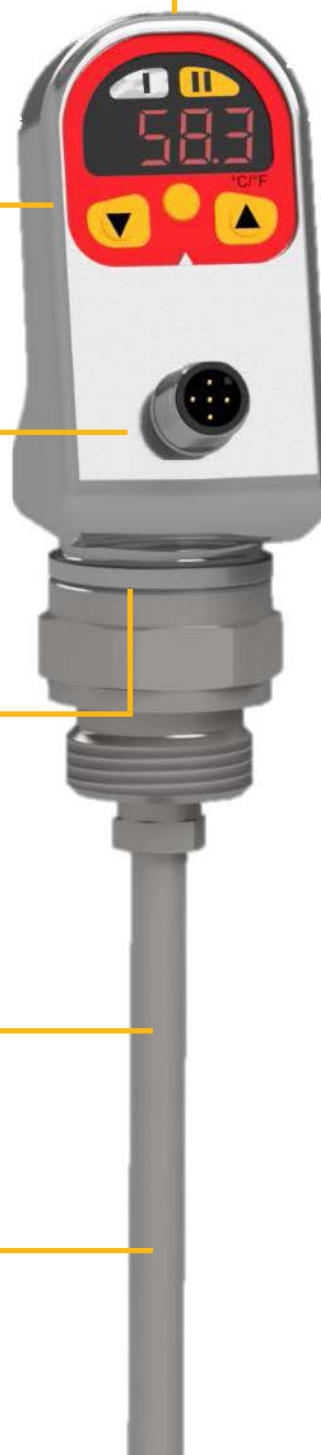
- Metallic housing
- High quality surface
- 240° rotatable
- IP67 protection

Measuring element

- Fixed PT1000 element
- Measuring range between -20°C to 85°C

Temperature probe

- Temperature probe in following length 250, 370 and 520mm
- High accuracy up to 0,35% FS
- Long-term stable measuring probe



SCTSDI Temperature Switch

Technical data

Ordercode temperature switch	SCTSDI-XXX-10-05		
Measuring range temperature	-20... 85		
Length (xxx in mm)	250	370	520
Lowest Reset point RSP	-20		
Highest Switching Point SP	85		
Display resolution	0,1		
Min. adjustable difference SP and rSP	1		

All units in bar

Process connection	
Process connection	G3/4 external
Tightening torque	105 Nm

Output variables	
Strom	0..20mA or 4..20mA (switchable)
Switching point	
Switching current	max. 200 mA
Short circuit current	400mA (short term), short circuit proof
Switching voltage	Supply voltage - 0,5 VDC
Execution	PNP/ NPN (programmable)
Max switching frequency	250 Hz

Accuracy (temperature)	
@ 25°C	±0,35% FS ±1 Digit
@ -20...+150°C	±1,5% FS ±1 Digit
Response speed	
Switching output	≤3 ms
Analogue output	≤3 ms
Sample rate	≤3 ms

Electrical connection	
Supply voltage V+	18...30 VDC
Connector	M12x1; 5pin
Short circuit protection	Yes
Reverse polarity protection	Yes
Overload protection	Yes
Overvoltage	50 V
Current consumption	<40 mA @24V
Inrush current	<100 mA @24V

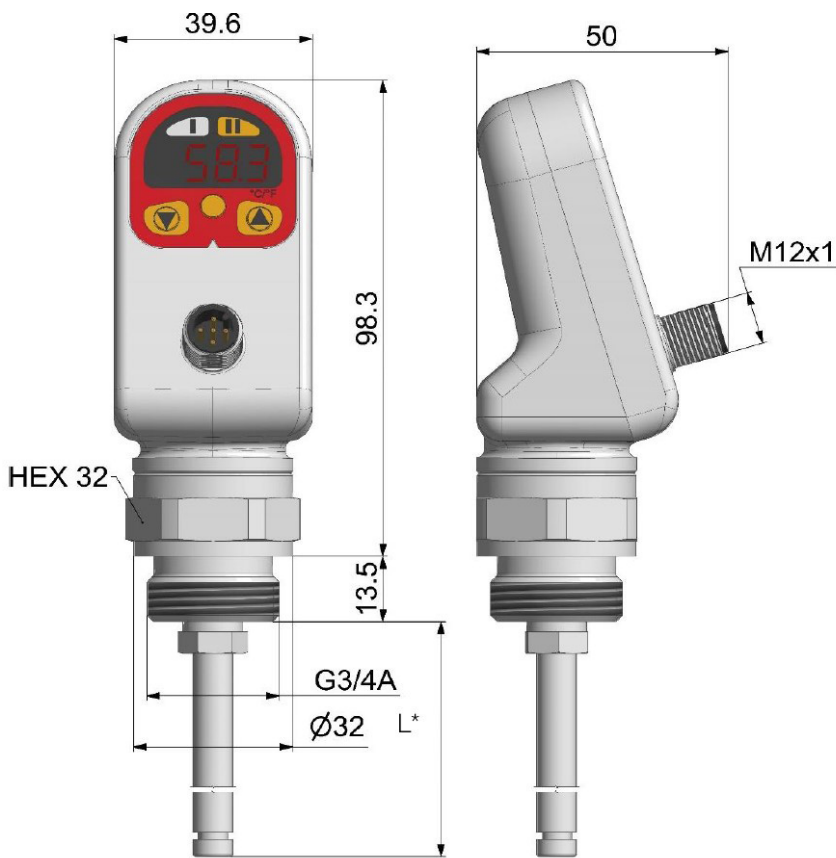
IO-Link	
Profile	Smart Sensor Profile 2nd Edition V1.1.2
Prozess Zykluszeit	4 ms
SIO-Mode	Yes
Master port type	A
Process data analogue	2 Byte Process data
	1 Byte Scaling factor
Process data binary	1 Byte
SDCI Standard	IEC 61131-9
Vendor ID	271 (dec) / 10f (hex)
Device IODD	https://parker.com/

SCTSDI Temperature Switch

Technical data

Housing	
Rotatability	>240°
Readability display	Reversal of viewing direction by 180° (programmable)
Display	7-Segment-LED (4-digit)
	2-color switching status display (red/ yellow)
Buttons	Digit height : 8 mm
	3 tactile buttons for up, down, enter
Material	
Housing	Zinc die-cast (bright nickel-plated)
Display cover	Polyester Autotex V207
Protection class	IP67
Weight	300 g
General	
MTTFd	>100 years
RoHS-Conformity	2011/65/EU, 2024/232

EM compatibility	
Immunity to interference	
Emission of interference	EN 55011
ESD	EN 61000-4-2
high-frequency electromagnetic fields	EN 61000-4-3
fast transient electrical disturbances	EN 61000-4-4
Conducted disturbances induced by high-frequency fields	EN 61000-4-6
Environmental conditions	
Ambient temperature	-20...+85 °C
Operating temperature	-20...+85 °C
Storage temperature	-40...+85 °C
Medium temperature	-20...+85 °C
Vibration resistance	DIN EN 60068-2-6, 20 g
Shock resistance	DIN EN 60068-2-27, 50g



L* = Probelength

SCTSDI Temperature Switch

Order code

SCTSDI Digital Temp Switch — **SCTSDI-250-10-05**
Probelength 250 mm
2 switching points; with current output
M12x1 connector; 5-Pin

SCTSDI Digital Temp Switch — **SCTSDI-370-10-05**
Probelength 370 mm
2 switching points; with current output
M12x1 connector; 5-Pin

SCTSDI Digital Temp Switch — **SCTSDI-520-10-05**
Probelength 520 mm
2 switching points; with current output
M12x1 connector; 5-Pin

Accessories

Securing clamp	SCSD-S24
Reducing adapter M22x1,5	RI22X1.5EDX1/4CF
Reducing adapter G1/2 BSPP	RI1/2EDX1/4CF
Attenuation adapter	SCA-1/4EDX1/4-D

Connection cable and single plug

Connection cable, assembled
(open cable end)

SCK-400 - xx - xx

Cable lenght (m)

2 m

5 m

10 m

02

05

10

Connector

M12 cable jack; straight

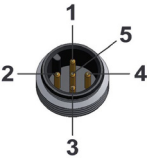
M12 cable jack; 90° angled

45

55

Single connector	
M12 cable jack; straight	SCK-145
M12 cable jack; 90° angled	SCK-155

PIN	SCxSDI-xxx-x4-x5
1	V+
2	S2 out
3	0 V/GND
4	S1 out / IO-Link
5	Analogue out



SCTSDI Modular Temperature Switch

Device features

Display

- Large displays for temperature
- 2-color switching status display
- Reversal of viewing direction by 180°
- Bevelled for better readability
- Units in °C and °F

Easy to use

- 3 tactile buttons
- VDMA-Menu navigation

Outputsignal

- IO-Link interface
- Stable metallic M12x1 thread
- Pin assignment according to VDMA
- With 2 switching outputs and switchable current output between 0..20 and 4..20mA

Housing

- Metallic housing
- High quality surface
- 240° rotatable
- IP67 protection

Measuring element

- M12x1 - 5Pin Connector
- Designed for an external PT1000 input Signal (temperature probe sold separately)
- Measuring range between -50°C and 150°C

Temperature probe

- Installation-space independent: The existing sensor probe can be connected via cable, regardless of location
- Direct connection via M12x1 connector is possible
- Separately available sensor probe: see SCT03



SCTSDI Modular Temperature Switch

Technical data

Ordercode temperature switch	SCTSDI-150-1T-05
Measuring range temperature	-50... 150
Lowest Reset point RSP	-50
Highest Switching Point SP	150
Display resolution	0,1
Min. adjustable difference SP and rSP	1

All units in bar

Process connection	
Process connection	M12x1 internal
Input signal	PT1000

Response speed	
Switching output	≤3 ms
Analogue output	≤3 ms
Sample rate	≤3 ms

Output variables	
Strom	0..20mA or 4..20mA (switchable)

Switching point	
Switching current	max. 200 mA
Short circuit current	400mA (short term), short circuit proof
Switching voltage	Supply voltage - 0,5 VDC
Execution	PNP/ NPN (programmable)
Max switching frequency	250 Hz

Accuracy (temperature)	
@ 25°C	±0,35% FS ±1 Digit
@ -20..+150°C	±1,5% FS ±1 Digit

Electrical connection	
Supply voltage V ₊	18...30 VDC
Connector	M12x1; 5pin
Short circuit protection	Yes
Reverse polarity protection	Yes
Overload protection	Yes
Overvoltage	50 V
Current consumption	<40 mA @24V
Inrush current	<100 mA @24V

IO-Link	
Profile	Smart Sensor Profile 2nd Edition V1.1.2
Prozess Zykluszeit	4 ms
SIO-Mode	Yes
Master port type	A
Process data analogue	2 Byte Process data 1 Byte Scaling factor
Process data binary	1 Byte
SDCI Standard	IEC 61131-9
Vendor ID	271 (dec) / 10f (hex)
Device IODD	https://parker.com/

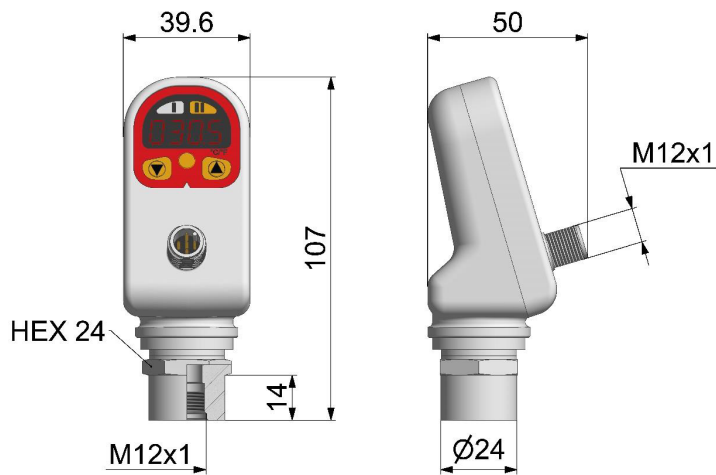
SCTSDI Modular Temperature Switch

Technical data

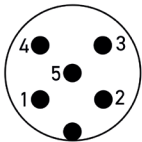
Housing		EM-compatibility	
Rotatability	>240°	Immunity to interference	
Readability display	Reversal of viewing direction by 180° (programmable)	Emission of interference	EN 55011
Display	7-Segment-LED (4-digit)	ESD	EN 61000-4-2
	2-color switching status display (red/yellow)	high-frequency electromagnetic fields	EN 61000-4-3
Digit height : 8 mm		fast transient electrical disturbances	EN 61000-4-4
Buttons	3 tactile buttons for up, down, enter	Conducted disturbances induced by high-frequency fields	EN 61000-4-6
Material		Environmental conditions	
Housing	Zinc die-cast (bright nickel-plated)	Ambient temperature	-20...+85 °C
Display cover	Polyester Autotex V207	Operating temperature	-20...+85 °C
Protection class	IP67	Storage temperature	-40...+85 °C
Weight	300 g	Medium temperature	-20...+85 °C
General		Vibration resistance	DIN EN 60068-2-6, 20 g
MTTFd	>100 years	Shock resistance	DIN EN 60068-2-27, 50g
RoHS-Conformity	2011/65/EU, 2024/232		

Compatible temperature probe	SCT03-007-Tx-07
PT1000	Class A
Accuracy	$\pm(0,15K+0,002 \times t)^{*1}$

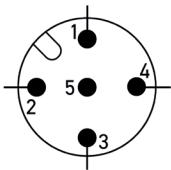
*1 t = measured temperature in °C (DIN EN 60751)
More details: See chapter SCT03 - Catalogue 4084



PIN	Assignment SCTSDI-150-1T-05 (Plug on Top)
1	V+
2	S2 out
3	0 V/GND
4	S1 out / IO-Link
5	Analogue out



PIN	Assignment SCTSDI-150-1T-05 (PT1000 socket below)
1	PT1000
2	-
3	PT1000
4	-
5	-



SCTSDI Modular Temperature Switch

Order code

SCTSDI Digital temperature switch
2 switching points; no analogue output **SCTSDI-150-1T-05**
M12x1 connector; 5-Pin

Accessory
Securing clamp **SCSD-S24**
Reducing adapter M22x1,5 **RI22X1.5EDX1/4CF**
Reducing adapter G1/2 BSPP **SCA-1/4-ED-1/2-ED**
Attenuation adapter **SCA-1/4EDX1/4-D**
PT1000 temperature probe **SCT03** (see chapter SCT03 - catalogue 4083)

Connection cable and single plug

Connection cable, assembled (open cable end)	SCK-400-xx-xx
Cable length (m)	
2 m	02
5 m	05
10 m	10
Connector	
M12 cable jack; straight	45
M12 cable jack; 90° angled	55

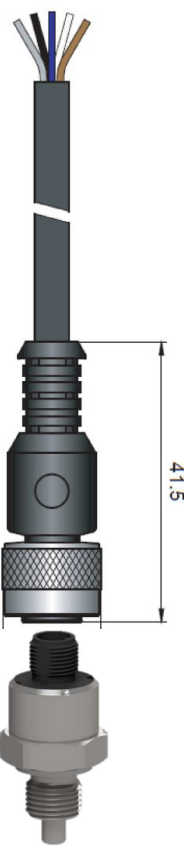
Single connector
M12 cable jack; straight **SCK-145**
M12 cable jack; 90° angled **SCK-155**
Adapter connector M12x1 **SCK-401-4M**

SCTSDI Modular Temperature Switch

Functional description



SCK-401-4M
(Adapter connector)



SCK-400-xx-xx
(Connection cable)

PT1000 from customer application

SCT03 (see chapter SCT03
catalogue 4083)

SCLSDI Level Switch

Device features

Display

- Large Display for level
- 2-color switching status display
- Reversal of viewing direction by 180°
- Beveled for better readability
- Units in mm, cm, m, inch und %

Easy to use

- 3 tactile buttons
- VDMA-Menu navigation

Outputsignal

- IO-Link interface
- Stable metallic M12x1 thread
- Pin assignment according to VDMA
- Can be selected between switching output, current output and voltage output

Housing

- Metallic housing
- High quality surface
- 240° rotatable
- IP67 protection

Installation

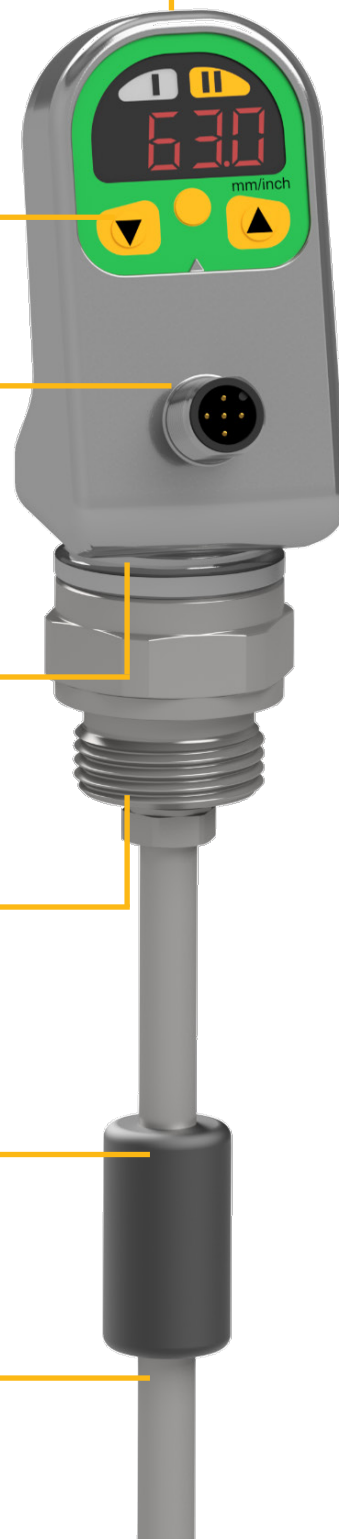
- G3/4" external thread
- Stainless Steel 1.4404
- One coupling hole

Float element

- Proven measuring system
- High float dynamics
- Small design
- Universal usage

Measuring element

- Measuring range up to 1000mm
- High Pressure rating up to 5bar
- High accuracy up to 0,5% FS
- Long-term stable measuring cell



SCLSDI Level Switch

Technical data

SCLSDI-	250	370	520	800	1000
Measuring range	30...215	30...335	30...485	20...750	20...950
Lowest reset point RSP	30	30	30	20	20
Highest switching point SP	215	335	485	750	950
Display resolution	5	5	5	10	10
Display resolution SP and rSP	5	5	5	10	10

All units in mm

Process connection

Process connection	G3/4 BSPP external
Tightening torque	165 Nm

Parts in contact with media

External thread	Stainless Steel: 1.4404 Sealing: ED NBR SH70
Sliding tube	Stainless steel 1.4301
Float element	NPR
Circlip	Bronze

Output variables

Current input	0..20mA or 4..20mA (programmable)
Voltage input	0..10V

Switching point

Switching current	max. 200 mA
Short circuit current	400mA (short term), short circuit proof
Switching voltage	Supply voltage - 0,5 VDC
Execution	PNP/ NPN (programmable)
Max switching frequency	250 Hz

Accuracy (level)

@ 25°C	±0,5% FS ±1 Digit
@ -20...+85°C	±1,5% FS ±1 Digit

Electrical connection

Supply voltage V ₊	18...30 VDC
Connector	M12x1; 4pol / M12x1; 5pol
Short circuit protection	Yes
Reverse polarity protection	Yes
Overload protection	Yes
Overvoltage	50 V
Current consumption	<40 mA @24V
Inrush current	<100 mA @24V

IO-Link

Profile:	Smart Sensor Profile 2nd Edition V1.1.2
Process Cyclustime	4 ms
SIO-Mode	Yes
Master port type	A
Process data analogue	2 Byte Process data 1 Byte Scaling factor
Process data binary	1 Byte
SDCI Standard	IEC 61131-9
Vendor ID	271 (dec) / 10f (hex)
Device IODD	https://parker.com/

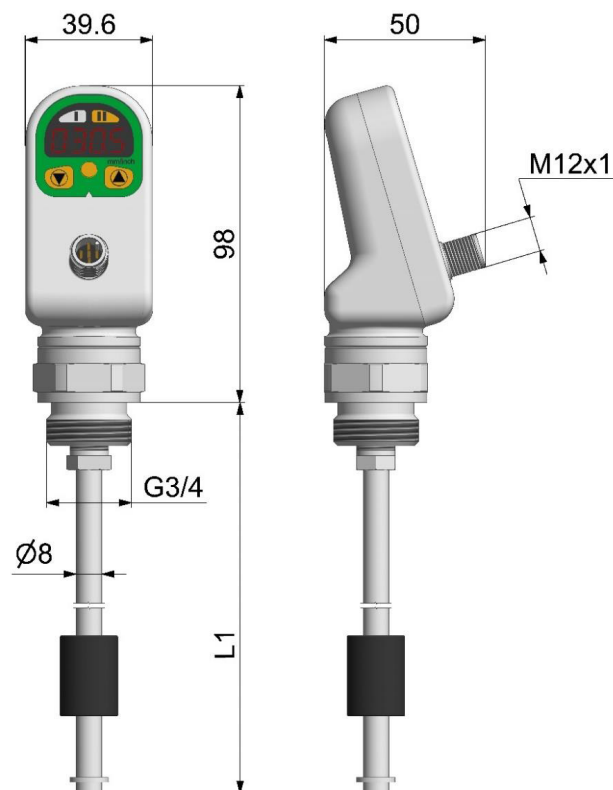
Response speed

Switching output	≤3 ms
Analogue output	≤3 ms
Sample rate	≤3 ms

SCLSDI Level Switch

Technical data

Housing		EM compatibility	
Rotatability	>240°	Immunity to interference	
Readability display	Reversal of viewing direction by 180° (programmable)	Emission of interference	EN 55011
Display	7-Segment-LED (4-digit)	ESD	EN 61000-4-2
	2-color switching status display (red/yellow)	high-frequency electro-magnetic fields	EN 61000-4-3
Buttons	3 tactile buttons for up, down, enter	fast transient electrical disturbances	EN 61000-4-4
Material		Conducted disturbances induced by high-frequency fields	EN 61000-4-6
Housing	Zinc die-cast (bright nickel-plated)	Environmental conditions	
Display cover	Polyester Autotex V207	Ambient temperature	-20...+85 °C
Protection class	IP67	Operating temperature	-20...+85 °C
Weight	~470 g	Storage temperature	-40...+85 °C
General		Medium temperature	-20...+85 °C
MTTFd	>100 years	Vibration resistance	DIN EN 60068-2-6, 20 g
RoHS-Conformity	2011/65/EU, 2024/232	Shock resistance	DIN EN 60068-2-27, 50g
Float dimensions			
M. Range<= 520mm	Ø17,8mm / length = 32mm		
M. Range<= 520mm	Ø21mm/ length = 36mm		



SCLSDI Level Switch

Order code

SCLSDI Digital Level switch

2 switching points; no analogue output SCLSDI-xxx-00-07
M12x1 connector; 4-Pin

1 switching point; with current output SCLSDI-xxx-10-07
M12x1 connector; 4-Pin

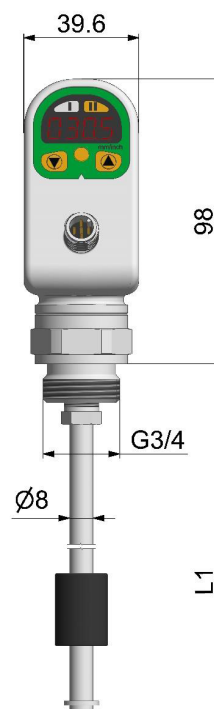
2 switching points; with current output or voltage output SCLSDI-xxx-x0-05
M12x1 connector; 5-Pin

Installation length L1

250 mm	250
370 mm	370
520 mm	520
800 mm	800
1000 mm	1000

Analogue output

No analogue output	0
0..20 mA / 4..20 mA	1
0..10 V	4



Accessory

Securing clamp
Reducing adapter M22x1,5
Reducing adapter G1/2 BSPP
Attenuation adapter

SCSD-S24
RI22X1.5EDX1/4CF
RI1/2EDX1/4CF
SCA-1/4EDX1/4-D

Connection cable and single plug

Connection cable, assembled SCK-400-xx-xx
(open cable end)

Cable length (m)

2 m	02
5 m	05
10 m	10

Connecting plug

M12 cable jack; straight	45
M12 cable jack; 90° angled	55

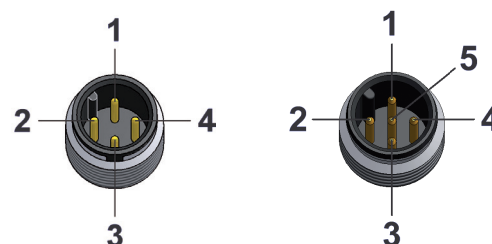
Single connector

M12 cable jack; straight SCK-145
M12 cable jack; 90° angled SCK-155

PIN	SCxSDI-xxx-x4-x7
1	V ₊
2	Analogue out
3	0 V/GND
4	S1 out / IO-Link

PIN	SCxSDI-xxx-04-x7
1	V ₊
2	S2 out
3	0 V / GND
4	S1 out / IO-Link

PIN	SCxSDI-xxx-x4-x5
1	V ₊
2	S2 out
3	0 V/GND
4	S1 out / IO-Link
5	Analogue out



SCLTSDI Level/Temperature Switch

Device features

Display

- Large displays for level and temperature
- 2-color switching status display
- Reversal of viewing direction by 180°
- Bevelled for better readability
- Units:
 - Level = mm, cm, m, inch and %
 - Temp = °C and °F

Easy to use

- 3 tactile buttons
- VDMA-Menu navigation

Outputsignal

- IO-Link interface
- Stable metallic M12x1 thread
- Pin assignment according to VDMA
- Can be selected between switching output, current output and voltage output

Housing

- Metallic housing
- High quality surface
- 240° rotatable
- IP67 protection

Installation

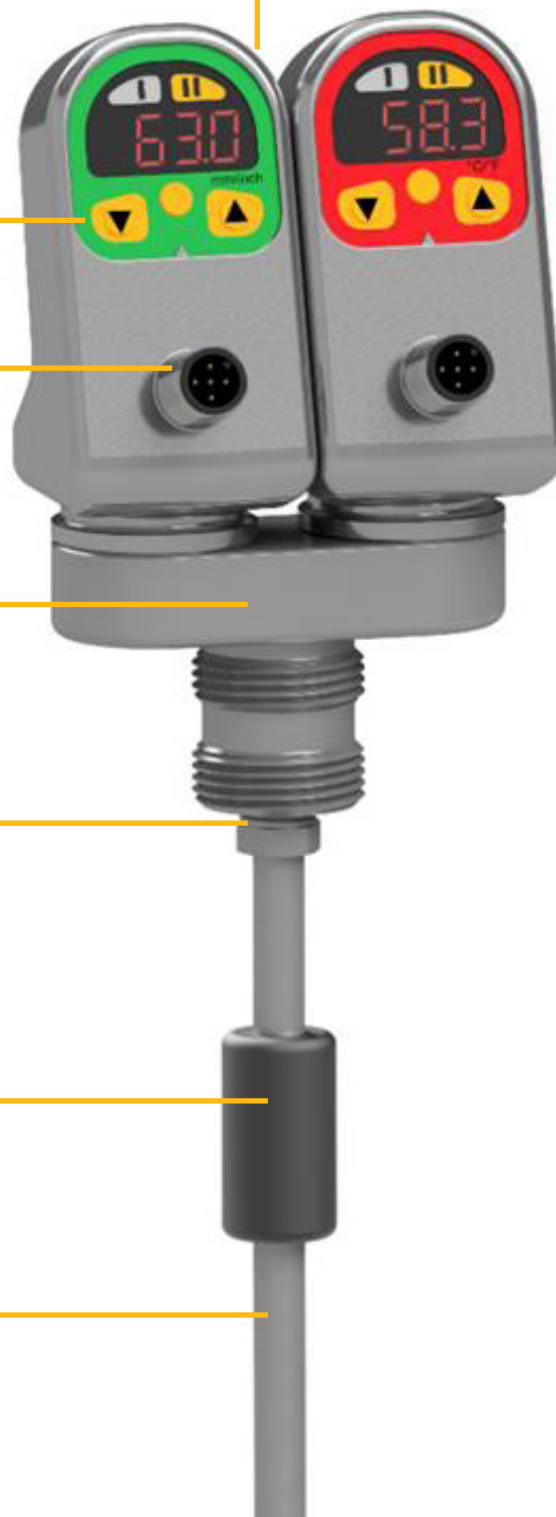
- G3/4" external thread
- Brass: CW724R
- 2 in 1 concept
- One coupling hole

Float element

- Proven measuring system
- High float dynamics
- Small design
- Universal usage

Measuring element

- Measuring range up to 1000mm and 85°C
- Tempsensor element integrated in the rod end
- High Pressure rating up to 5 bar
- High accuracy up to 0,5% FS
- Long-term stable measuring cell



SCLTSDI Level/Temperature Switch

Technical data

SCLTSDI-	250	370	520	800	1000
Measuring range	30...215	30...335	30...485	20...750	20...950
Lowest reset point RSP	30	30	30	20	20
Highest switching point SP	215	335	485	750	950
Display resolution	5	5	5	10	10
Display resolution SP und rSP	5	5	5	10	10
Measuring temperature range	-20°C .. 85°C	-20°C .. 85°C	-20°C .. 85°C	-20°C .. 85°C	-20°C .. 85°C

All units in mm

Process connection

Process connection	G3/4 BSPP external
Tightening torque	105 Nm

Parts in contact with media

External thread	Brass: CW724R (CuZn21Si3P) Sealing: ED NBR SH70
Sliding tube	Stainless steel 1.4301
Float element	NBR
Circlip	Bronze

Output variables

Current input	0..20mA or 4..20mA (programmable)
Voltage input	0..10V

Switching point

Switching current:	max. 200 mA
Short circuit current:	400mA (short term), short circuit proof
Switching voltage:	Supply voltage - 0,5 VDC
Execution:	PNP/ NPN (programmable)
Max switching frequency	250 Hz

Accuracy (level)

@ 25°C	±0,50% FS ±1 Digit
@ -20...+85C	±1,5% FS ±1 Digit

Accuracy (temperature)

@ 25°C	±0,35% FS ±1 Digit
@ -20...+85C	±1,5% FS ±1 Digit

Electrical connection

Supply voltage V _s	18...30 VDC
Connector	M12x1; 4pol / M12x1; 5pol
Short circuit protection	Yes
Reverse polarity protection	Yes
Overload protection	Yes
Overvoltage	50 V
Current consumption	<40 mA @24V
Inrush current	<100 mA @24V

IO-Link

Profile:	Smart Sensor Profile 2nd Edition V1.1.2
Process Cyklustime	4 ms
SIO-Mode	Yes
Master port type	A
Process data analogue	2 Byte Process data 1 Byte Scaling factor
Process data binary	1 Byte
SDCI Standard	IEC 61131-9
Vendor ID	271 (dec) / 10f (hex)
Device IODD	https://parker.com/

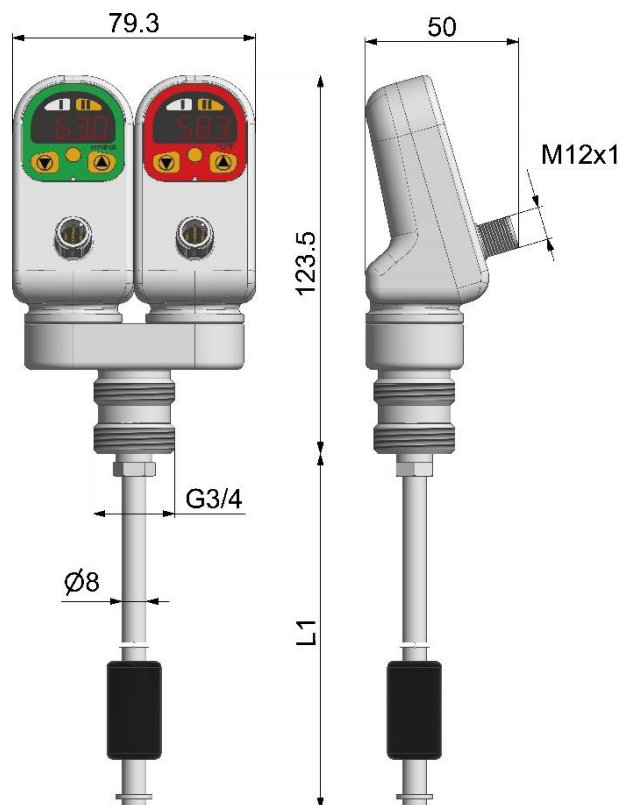
Response speed

Switching output	≤3 ms
Analogue output	≤3 ms
Sample rate	≤3 ms

SCLTSDI Level/Temperature Switch

Technical data

Housing		EM compatibility	
Rotatability	>240°	Immunity to interference	
Readability display	Reversal of viewing direction by 180° (programmable)	Emission of interference	EN 55011
Display	7-Segment-LED (4-digit)	ESD	EN 61000-4-2
	2-color switching status display (red/yellow)	high-frequency electromagnetic fields	EN 61000-4-3
Buttons	3 tactile buttons for up, down, enter	fast transient electrical disturbances	EN 61000-4-4
Material		Conducted disturbances induced by high-frequency fields	EN 61000-4-6
Housing	Zinc die-cast (bright nickel-plated)	Environmental conditions	
Display cover	Polyester Autotex V207	Ambient temperature	-20...+85 °C
Protection class	IP67	Operating temperature	-20...+85 °C
Weight	~860 g	Storage temperature	-40...+85 °C
General		Medium temperature	-20...+85 °C
MTTFd	>100 years	Vibration resistance	DIN EN 60068-2-6, 20 g
RoHS-Conformity	2011/65/EU, 2024/232	Shock resistance	DIN EN 60068-2-27, 50g
Float dimensions			
M. Range<= 520mm	Ø17,8mm / length = 32mm		
M. Range<= 520mm	Ø21mm/ length = 36mm		



SCLTSDI Level/Temperature Switch

Order code

SCLTSDI Digital Level/Temp switch

2 switching points; no analogue output SCLTSDI-xxx-00-07
M12x1 connector; 4-Pin

1 switching point; with current output SCLTSDI-xxx-10-07
M12x1 connector; 4-Pin

2 switching points; with current output or voltage output SCLTSDI-xxx-x0-05
M12x1 connector; 5-Pin

Installation length L1 mm

250 mm	250
370 mm	370
520 mm	520
800 mm	800
1000 mm	1000

Analogue output

No analogue output	0
0..20 mA / 4..20 mA	1
0..10 V	4

Accessory

Securing clamp
Reducing adapter M22x1,5
Reducing adapter G1/2 BSPP
Attenuation adapter

SCSD-S24
RI22X1.5EDX1/4CF
RI1/2EDX1/4CF
SCA-1/4EDX1/4-D

Connection cable and single plug

Connection cable, assembled

(open cable end)

SCK-400-xx-xx

Cable length (m)

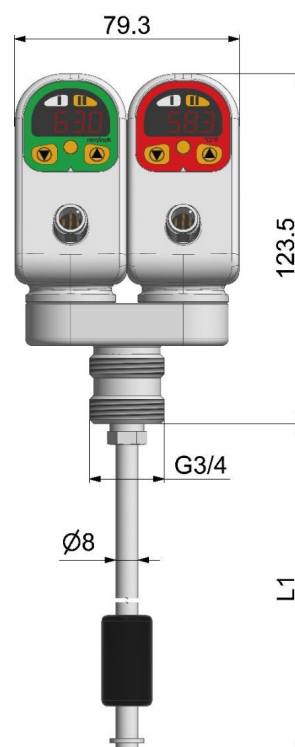
2 m	02
5 m	05
10 m	10

Connector

M12 cable jack; straight	45
M12 cable jack; 90° angled	55

Single connector

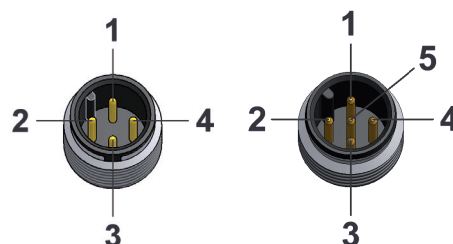
M12 cable jack; straight SCK-145
M12 cable jack; 90° angled SCK-155



PIN	SCxSDI-xxx-x4-x7
1	V ₊
2	Analogue out
3	0 V/GND
4	S1 out / IO-Link

PIN	SCxSDI-xxx-04-x7
1	V ₊
2	S2 out
3	0 V/GND
4	S1 out / IO-Link

PIN	SCxSDI-xxx-x4-x5
1	V ₊
2	S2 out
3	0 V/GND
4	S1 out / IO-Link
5	Analogue out



SCK cable

Device features

- One cable for all
- Compact size
- Interference-free
- Compatible to:
 - Sensors
 - Controllers
- M12 plug
- DIN EN 175301 (Device plug)
- Available in a variety of lengths



The **SensoControl®** cable was designed for use with the industrial sensors and switches.

Thus the M12 cable and M12 plug are

- Compact
- Shielded
- Five-pole

5-pole version

The 5-pole cable is suitable for both 4-pole and 5-pole connections. The sensor variants with a 4-pole connector are fully compatible with the 5-pole cable.

So despite different pin counts on the pressures switch (Controller Family SCxSD and SCOTC) and sensors, it is always possible to use just one cable version (5-pole) regardless of the plug version.

The SCK-400-xxx-x5 cables fit to all components in this catalogue using M12 connectors.

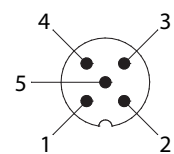
Shielding

Shielding protects against interference and ensures improved operational safety.

- Higher EMC protection

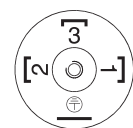
Pin assignment

SCK-400-xx-x5



PIN			
1	bn	brown	braun
2	wh	white	weiß
3	bu	blue	blau
4	bk	black	schwarz
5	gy	grey	grau

SCK-400-xx-56

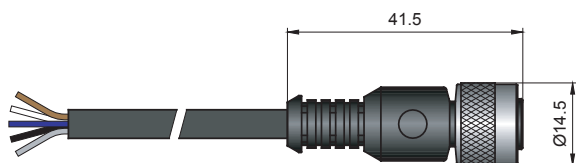


PIN			
1	ye	yellow	gelb
2	gn	green	grün
3	bn	brown	braun

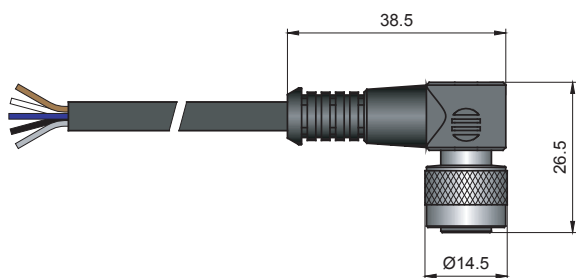
SCK cable

Connection cable

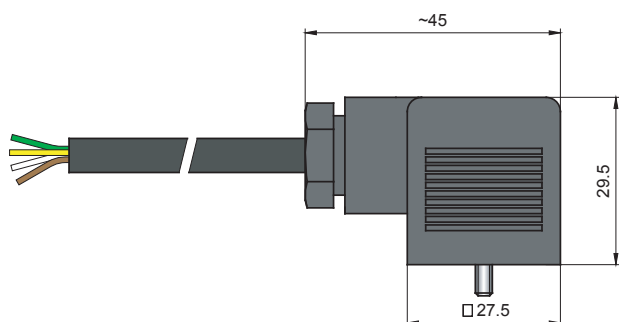
SCK-400-xx-45



SCK-400-xx-55

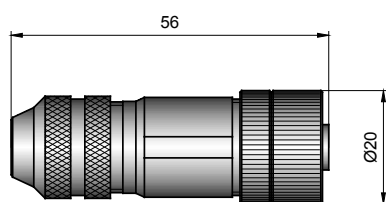


SCK-400-xx-56

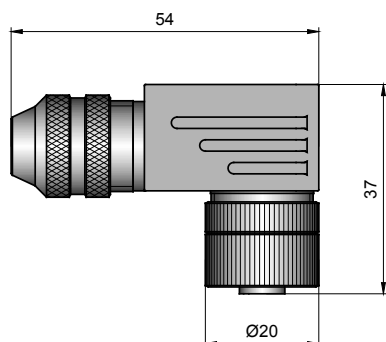


Single connector

SCK-145

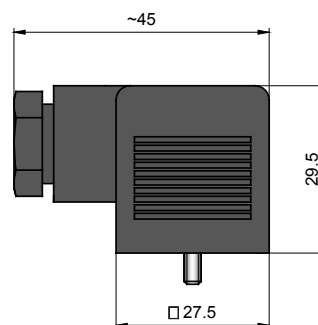


SCK-155



Single connector

SCK-006 (Device plug)



Connection cable and single plug

Connection cable, assembled

(open cable end)

Cable length (m)

2 m	02
5 m	05
10 m	10

Connecting plug

M12 cable jack; straight	45
M12 cable jack; 90° angled	55
Cable socket DIN EN 175301-803 Form A (old DIN 43650)	56

Single connector

M12 cable jack; straight
M12 cable jack; 90° angled
Cable socket DIN EN 175301-803 Form A (old DIN 43650)

SCK-145
SCK-155
SCK-006

SCA adapter

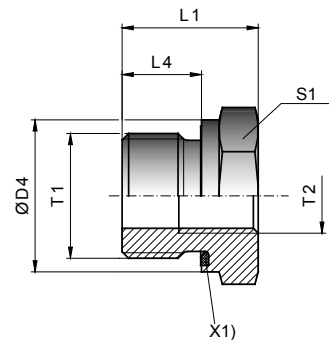
SCA-1/4 reduction adapter

The SCA-1/4 provides compatibility for earlier sensor versions with the hydraulic connection M22x1.5 or G1/2 BSPP.

- When replacing earlier versions

This allows facilities to be updated without major planning overhead.

SCA-1/4-M22x1.5-ED
SCA-1/4-ED-1/2-ED



X1) EOLASTIC-seal

	T1	T2	ØD4	L1	L4	S1	Weight (g/1 St)	PN (bar) ¹⁾	DF ^{**}
SCA-1/4-M22x1.5-ED	M22x1.5	G1/4 BSPP	27	24	14	27	56	400	4
SCA-1/4ED1/2-ED	G1/2 BSPP	G1/4 BSPP	27	24	14	27	56	400	4

SCA-1/4 attenuation adapter

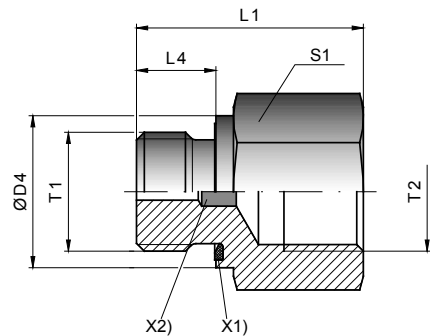
System-related pressure spikes are reduced with the SCA-1/4-EDX-1/4-D.

- Attenuation for pressure peaks

The G1/2 BSPP version ensures compatibility for earlier sensor versions to the G1/2 BSPP hydraulic connection.

- When replacing earlier versions

SCA-1/4-EDX-1/4-D



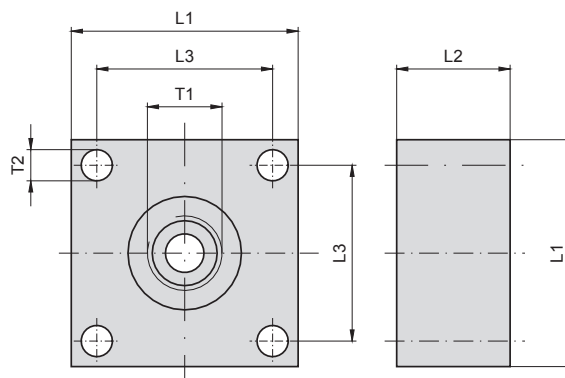
X1) EOLASTIC-seal

	T1	T2	ØD4	L1	L4	S1	Weight (g/1 St)	PN (bar) ¹⁾	DF ^{**}
SCA-1/4EDX1/4-D	G1/4A BSPP	G1/4 BSPP	19	34	12	22	61	630	3.5

SCA adapter

SCPSD flange adapter SCAF-1/4-40 for mechanical pressure switch

When replacing existing mechanical pressures switches SCAF-1/4-40
with 40x40mm flange connections



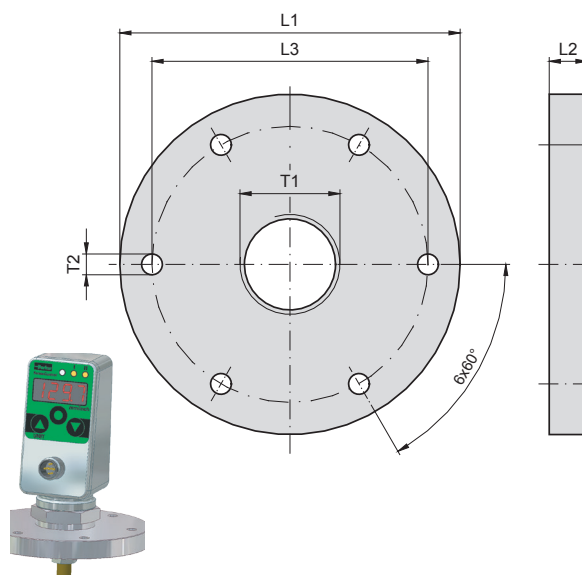
SCAF-1/4-40
for mechanical pressure switch

SCAF-1/4-40

T1	T2	L1	L2	L3	Weight (g/1 St)	PN (bar) ¹⁾ Alu	DF **
G1/4 BSPP	5.5	40	20	31	15	400	4

SCLSD/SCLTSD flange adapter SCAF-3/4-90 6-hole connection DIN 24557, part 2

For LevelController and LevelTemp Controller (SCLSD and SCLTSD), a compatibility to the tank connections 6-hole DIN 24557, part 2, is ensured.



SCAF-3/4-90
6-hole connection DIN 24557, part 2

SCAF-3/4-90

T1	T2	L1	L2	L3	Weight (g/1 St)	Material
G3/4 BSPP	5.5	90	10	73	520	Stainless steel

** DF = Design Factor (safety factor)

Installation and safety instructions



The CE mark indicates a high-quality device that complies with the European directive 89/336/EEG and EMVG.

We confirm that these products comply with the following standards:

EMC

- Electromagnetic emission: EN 61000-6-3
- Electromagnetic immunity: EN 61000-6-2

Important

- Electromagnetic disturbances can affect the desired signal.
- Apply all general EMC strategies when planning facilities and machines.
- We recommend using shielded cables (SCK-400-xx-x5) in order to achieve better EMC immunity.
- Make sure you route analogue and data cables so that there is a sufficient gap between them.
- An effective earthing strategy will help you to avoid measuring errors.

Always connect metal housings with the reference ground. The PE protective earth should have a low-ohm connection. According to VDE 0701, the PE resistance must be measured.

Power feed voltage



Each sensor series specifies the recommended feed voltage to be used when operating the standard sensor. We recommend using a low-noise, high-quality, constant voltage source. Certain specifications (such as sensitivity and thermal sensitivity shift) may change when other power feeds are used. Each sensor is trimmed to its peak performance. The sensor's performance may change when other power feed types are used. Make sure you comply with the polarity and earthing regulations.

Improperly connected feed wires can damage sensors and amplifiers!

If one pole of the sensor feed is automatically earthed via the sensor's processing system, then you should avoid an additional earth on the sensor signal wire. This would cause the sensor to short circuit and damage the sensor.

Do not apply feed-in voltage to the output wires. This will permanently damage the sensors!



The sensor will be damaged if the data sheet specifications and maximum recommended feed voltage levels are exceeded!

Compatibility with media (substances)

SensoControl® products which come into contact with the substance are not produced in an oil-free or fat-free environment.

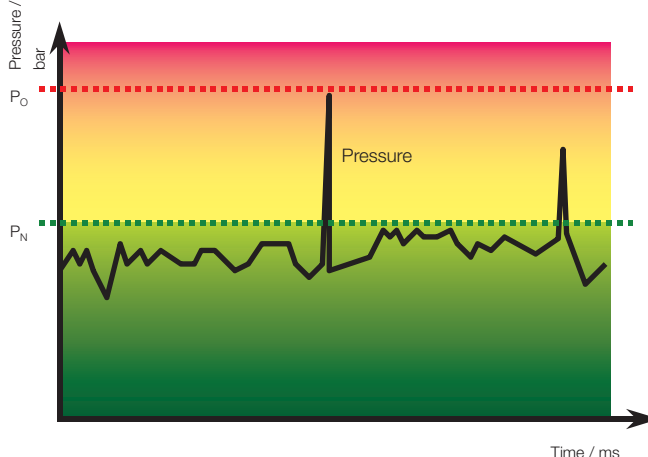
Therefore these products are **not** suitable for use in applications which use explosive mixtures of oil and gas (e.g. oxygen or compression). This could lead to a danger of explosion!

Danger of explosion!

Only use substances which are compatible with the components that come into contact with the substance. (Refer to the data sheets)

Please consult with the plant manufacturer or the manufacturer of the substance if you have any questions. (Refer to catalogue 4100 chapter C).

Pressure range selection



When selecting pressure components, ensure that the overload pressure P_{max} will not be exceeded.

It is possible that the pressure cell can be deformed when the overload pressure P_{max} is exceeded (depending on the duration, frequency and level of the pressure spike).

Note: The "diesel effect" caused by entrapped air can result in pressure spikes that far exceed the maximum pressure.

The nominal pressure P_N of the pressure component (sensor/switch) should be higher than the nominal pressure of the system to be measured.

Appendix

Temperature conversion table

Celsius to Fahrenheit

°C	°F
150	302
145	293
140	284
135	275
130	266
125	257
120	248
115	239
110	230
105	221
100	212
95	203
90	194
85	185
80	176
75	167
70	158
65	149
60	140
55	131
50	122
45	113
40	104
35	95
30	86
25	77
20	68
15	59
10	50
5	41
0	32
-5	23
-10	14
-15	5
-20	-4
-25	-13
-30	-22
-35	-31
-40	-40
-45	-49
-50	-58

Fahrenheit to celsius

°F	°C
340	171
330	166
320	160
310	154
300	149
290	143
280	138
270	132
260	127
250	121
240	116
230	110
220	104
210	99
200	93
190	88
180	82
170	77
160	71
150	66
140	60
130	54
120	49
110	43
100	38
90	32
80	27
70	21
60	16
50	10
40	4
30	-1
20	-7
10	-12
0	-18
-10	-23
-20	-29
-30	-34
-40	-40
-50	-46
-60	-51

Pressure conversion table

bar to psi

bar	psi
1000	14505
800	11604
600	8703
500	7253
400	5802
250	3626
160	2321
100	1451
60	870
40	580
35	508
25	363
16	232
10	145
6	87
4	58
2.5	36
1.6	23
1	15

psi to bar

psi	bar
10000	689
9000	620
7000	483
6000	414
4000	276
3000	207
2500	172
1000	69
900	62
600	41
500	34
400	28
250	17
150	10.3
100	6.9
90	6.2
60	4.1
40	2.8
25	1.7
10	0.7

Examples

Temperature conversion

Initial value: 100

°C in °F: 212 °F

°F in °C: 37.78 °C

Pressure conversion

Initial value: 35

bar in psi: 507.675 psi

psi in bar: 2.41296 bar

Index

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SC-910	37	SCPSi	62-65
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SCK-155	84		
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SCLTSD-...	79-82		
SCOTC-...	72-77		
SCP03-...	12-16		
SCP04-...	15-19		
SCP08	20-21		
SCP09	22-26		
SCP10	27-31		
SCP11	32-35		

Old and new references

Old order number	New order number	Old order number	New order number
SCK-007	SCK-145	SCP-xxx-x4-0x-MO	SCP03-xxx-x4-0x
SCK-045	SCK-145	SCP-xxx-x4-0x	SCP03-xxx-x4-0x
SCK-047	SCK-145	SCP-xxx-10-06	SCP03-xxx-14-06 + SCA-1/4-M22x1.5-ED
SCK-055	SCK-155	SCP-xxx-10-07	SCP03-xxx-14-07 + SCA-1/4-M22x1.5-ED
SCK-057	SCK-155	SCP-xxx-12-06	SCP03-xxx-14-06 + SCA-1/4-ED-1/2-ED
SCK-147	SCK-145	SCP-xxx-12-07	SCP03-xxx-14-07 + SCA-1/4-ED-1/2-ED
SCK-157	SCK-155	SCP-xxx-20-06	SCP03-xxx-24-06 + SCA-1/4-M22x1.5-ED
SCK-200-xxx-45	SCK-400-xxx-45	SCP-xxx-20-07	SCP03-xxx-24-07 + SCA-1/4-M22x1.5-ED
SCK-200-xxx-47	SCK-400-xxx-45	SCP-xxx-22-06	SCP03-xxx-24-06 + SCA-1/4-ED-1/2-ED
SCK-200-xxx-55	SCK-400-..55	SCP-xxx-22-07	SCP03-xxx-24-07 + SCA-1/4-ED-1/2-ED
SCK-200-xxx-56	SCK400-xxx-56	SCP-xxx-30-06	SCP03-xxx-34-06 + SCA-1/4-M22x1.5-ED
SCK-200-xxx-57	SCK-400-..55	SCP-xxx-30-07	SCP03-xxx-24-07 + SCA-1/4-M22x1.5-ED
SCK-400-xxx-06	SCK-400-xxx-56	SCP-xxx-32-06	SCP03-xxx-34-06 + SCA-1/4-ED-1/2-ED
SCK-400-xxx-07	SCK-400-xxx-45	SCP-xxx-32-07	SCP03-xxx-24-07 + SCA-1/4-ED-1/2-ED
SCK-400-xxx-47	SCK-400-xxx-45	SCP-xxx-40-06	SCP03-xxx-44-06 + SCA-1/4-M22x1.5-ED
SCK-400-xxx-57	SCK-400-..55	SCP-xxx-40-07	SCP03-xxx-44-07 + SCA-1/4-M22x1.5-ED
SCPSD-xxx-04-05	SCPSD-xxx-04-17	SCP-xxx-42-06	SCP03-xxx-44-06 + SCA-1/4-ED-1/2-ED
SCPSD-xxx-04-06	SCPSD-xxx-04-16	SCP-xxx-42-07	SCP03-xxx-44-07 + SCA-1/4-ED-1/2-ED
SCPSD-xxx-04-07	SCPSD-xxx-04-17	SCP01	SCP03
SCPSD-xxx-14-05	SCPSD-xxx-14-15	SCP02	SCP03

Please ask about compatible products for non-listed items.

